

Review

An overview on multi-agent consensus under adversarial attacks[☆]Hideaki Ishii^{a,*}, Yuan Wang^b, Shuai Feng^c^a Department of Computer Science, Tokyo Institute of Technology, Yokohama, Japan^b Division of Decision and Control Systems, KTH Royal Institute of Technology, Stockholm, Sweden^c School of Automation, Nanjing University of Science and Technology, Nanjing, China

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ABSTRACT

This paper presents an overview on the recent advances in the research of security of cyber-physical systems. We place particular emphases on consensus problems for multi-agent systems in hostile environments and their analyses on the resiliency against two types of attacks. First, we discuss a class of data injection attacks by focusing on the approach based on mean subsequence reduced (MSR) algorithms and their variants. Agents equipped with such algorithms will ignore their neighbors taking extreme state values. Characterizations on the properties necessary for network topologies and moreover a number of extensions with enhanced resiliency will be established. As the second class of attacks, the effects of denial-of-service (DoS) attacks will be examined in the context of multi-agent consensus. By employing a DoS model based on the energy constraints of the attacker, we will observe that robustness against such attacks may depend on system properties such as dynamics of the individual agents and network structures. Applications of the algorithms will be further discussed for clock synchronization in wireless sensor networks and control of a group of mobile robots.

1. Introduction

In recent years, control systems have significantly changed in their structures and operations under the extensive utilization of information and communication technologies. As a consequence of the trend towards more connectivity among control components, large-scale systems today operate in a distributed fashion at different levels where controllers exchange information in real time over communication networks (Kim & Kumar, 2012). For example, via the extensive spreading of the Internet of Things, control related data from many sensors can be collected over wireless networks, which has enlarged the domain of control to critical social infrastructure including power systems and traffic systems.

Under these new developments, however, the risk of cyber attacks targeting control systems has tremendously increased. Communication channels as well as the devices used in control systems are vulnerable to adversarial cyber attacks. This is partly because they have critical real-time requirements, and encryption in communication is often not an option to keep time delays minimal. The potential faults and damages that may result from such attacks can physically take place, and thus security is of importance to the safety of not only the control systems themselves but also the society at large.

In response to the serious incidents that occurred in various parts of the world in the past decade, cyber security of control systems

has grown into a major research issue, and there is a large body of literature on the subject; see, for example, Cárdenas, Amin, and Sastry (2008), Dibaji et al. (2019), Ferrari and Teixeira (2021), Ishii and Zhu (2022), Mo et al. (2012), Pasqualetti, Dörfler, and Bullo (2013), Sandberg, Amin, and Johansson (2015) and the references therein. Different types of cyber attacks can threaten the operation of networked control systems. For instance, in a vulnerable network, measurement and control data can become targets for eavesdropping by adversarial attackers, leading to leakage of private and confidential information. Under more hostile environments, communication of such data can be more actively blocked and even falsified. Especially, if the attackers have sufficient knowledge about the system and its dynamics, false data injection attacks may be launched to the control inputs, resulting in manipulation of the internal states of the systems without being detected from the outputs by system operators.

In this paper, as the class of cyber-physical systems, our focus is on multi-agent systems and their distributed control and coordination (e.g., Bullo, 2018; Francis & Maggiore, 2007; Mesbahi & Egerstedt, 2010; Ren & Cao, 2011). Such systems have received much attention in various areas including control, robotics, communications, complex networks, and computer science. In large-scale multi-agent systems, consensus problems form one of the fundamental problems. There, agents interact locally and exchange their information with each other

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in order to arrive at the global objective of sharing a common state value. In hostile environments where frequent faults and adversarial attacks can be present, it is crucial to defend distributed consensus algorithms by ensuring their security so as to keep their decision makings from the influence of the uncertainties created by external attackers.

The objective of this paper is to provide an overview on different types of cyber attacks on consensus problems and the corresponding measures for secure and resilient control to mitigate their influence on the overall systems and networks. In the area of systems control, this is a relatively new subject of research, but the interest in the community has recently grown significantly. This is partly due to the breadth of studies on multi-agent consensus carried out over the last two decades. In principle, all of these studies can be extended with further consideration of security issues. At the same time, this line of research has interdisciplinary aspects and, in particular, has brought new interesting interactions with the field of computer science, which has a rich history of studies on security against faults as well as distributed algorithms (e.g., Lynch, 1996).

In particular, in this paper, we present secure and resilient protocols for multi-agent systems against different attacks. This is a very broad subject, and we limit our scope of the paper to recent works on two types of attacks: False data injections (FDI) and denial-of-service (DoS) attacks. These attacks are briefly explained as follows. (i) The first type, the FDI attacks, occurs when agents transmit false data to their neighbors. They may result from abnormal behaviors due to faults in devices of the agents or adversaries present in the network emitting unwanted packets. Such agents do not follow the given algorithms and might even attempt to keep the non-faulty, regular agents from reaching consensus. (ii) On the other hand, the second type, DoS attacks, can be classified as those based on jamming and DoS attacks, which can temporarily prevent communication over networks. Such attacks are relatively easy to launch and require little information about the agent systems. For example, malicious routers in a network may intentionally drop control data. Moreover, in wireless networks, an attacker can block the transmissions by simply emitting strong interference signals.

In what follows, we discuss consensus-based approaches under these types of attacks and related research. We will put certain stress on the historical aspects from the area of computer science.

1.1. Resilient consensus in the presence of malicious agents

Distributed consensus/agreement problems have been long studied in several areas. Early works can be found in statistics (Stone, 1961; Winkler, 1968), which are concerned with the problem of a group faced with a decision-making situation. When a decision maker has little knowledge with regard to the parameters of interest, he may decide to consult a number of experts and obtain their quantified judgments. The decision maker must combine the distributions assessed by the experts and form a single distribution. The problem of the determination of this single distribution can be seen as one of the first versions of *consensus* notions.

On the other hand, along with the development of fault-tolerant distributed computing, computer scientists started to study distributed consensus problems in the presence of faulty processes. The early works such as (Dolev, Lynch, Pinter, Stark, & Weihl, 1986) and (Pease, Shostak, & Lamport, 1980) studied the so-called Byzantine agreement problem. This problem requires that all non-faulty processes obtain identical voted values for any set of initial values. The basic principle in realizing resilience is simple and intuitive. It consists of two characteristic features: (i) To set a limit on the number of adversaries in the system and then (ii) to find the majority in the values of the neighbors and ignore the minority or the outliers. Meanwhile, this problem has found different real-world applications such as sensor data management and fault-tolerant clock synchronization (Lamport & Melliar-Smith, 1985; Schneider, 1987).

While many of these works deal with binary-valued states, efforts to extend to problems involving real-valued states have been pursued, which are sometimes referred to as *approximate Byzantine agreement* problems; see, for example, Dolev et al. (1986) and Kieckhafer and Azadmanesh (1992, 1994). In such problems, the non-faulty processes need not achieve exact agreement, and they only need to agree on a value within a specified tolerance. In particular, approximate Byzantine agreement protocols are characterized by two conditions:

- (i) (*Validity*): The voted value held by each non-faulty process is within the range of the initial values held by the non-faulty processes.
- (ii) (*Agreement*): The voting algorithms executed by all non-faulty processes eventually halt with voted values that are within a small constant ϵ of each other.

Distributed protocols that effectively guarantee these conditions are known under the name *mean subsequence reduced (MSR)* algorithms.

In the past decade, as security problems have become a critical issue in the domain of cyber-physical systems, Byzantine agreement and related problems have gained revived interests in a wide range of disciplines. This includes not only computer science, but also systems control, signal processing, robotics, and power systems. These studies are particularly motivated by the active research on multi-agent systems that have taken place since the mid-2000s, where security issues have not been commonly addressed. In dealing with cyber-physical systems, resilient approaches have been developed with consideration of complex dynamics within the individual agents as well as in the networks representing the interaction among the many agents.

In resilient consensus problems, key progresses have been made in the line of research initiated by the works (Dibaji, Ishii, & Tempo, 2018; LeBlanc, Zhang, Koutsoukos, & Sundaram, 2013; Vaidya, Tseng, & Liang, 2012). We must highlight several points that distinguish these works from the earlier ones:

1. Agents are now allowed to interact over *sparse networks*, as opposed to the all-to-all networks that have been common in previous works. This is natural from the perspective of multi-agent applications such as sensor networks and groups of autonomous vehicles whose wireless communication ranges may be limited.
2. Full characterizations have been obtained on connectivities in the underlying networks necessary for arriving at consensus. To this end, a new notion of graphs called *robustness* has been introduced.
3. When the communication among agents is subject to time delays, the agents can make updates in their states in a more real-time manner. This is in contrast to the methods based on the so-called *rounds*, which essentially mandate agents to update only when the same number of messages is received from all neighbors over time. Such a method can easily incur long delays in the consensus forming processes.

Results on resilient consensus based on MSR algorithms and related topics will be the main subject of the current paper. We will start with the initial problem formulation for the simplest cases and then present variations of the problem. They include agent systems with constraints in communication as well as agents and adversaries with further dynamics and different levels of resilience and attack capabilities, respectively.

1.2. Multi-agent consensus under DoS attacks

Next, we provide a brief overview on the recent research on another class of attacks, DoS attacks, and their influence on multi-agent consensus. This type of attacks can induce packet drops maliciously and hence corrupts the availability of data transmitted within multi-agent systems. In the presence of DoS attacks, such information exchange may not be possible, preventing the multi-agent systems from achieving their

control objectives. They can be easily launched especially in wireless communication, which is a common means for the interaction among agents in many emerging applications. Similar models can be employed to represent jamming attacks caused by strong interference signals in wireless channels.

Compared with random packet losses due to network congestions under normal conditions, communication failures induced by DoS attacks can exhibit temporal profiles quite different. Particularly, packet dropouts resulting from malicious DoS need not follow a given class of probability distributions (Amin, Cárdenas, & Sastry, 2009). Therefore, the analysis techniques relying on probabilistic arguments may not be applicable. This poses new challenges in theoretical studies and controller designs.

Recently, systems under such attacks have been studied from the viewpoint of systems control initially for centralized systems, e.g., by the works of Cetinkaya, Ishii, and Hayakawa (2019), De Persis and Tesi (2015), Feng, Cetinkaya, Ishii, Tesi, and De Persis (2021), Feng and Tesi (2017), Li, Quevedo, Dey, and Shi (2017) and Qin, Li, Shi, and Yu (2017). In De Persis and Tesi (2015), a framework has been introduced where DoS attacks are characterized by their levels of frequency and duration, determined by the energy available to the attacker; this framework is then utilized to find conditions for preserving stability of control systems through state feedback. For multi-agent systems under DoS attacks, there are recent developments for cooperative control such as consensus, output regulation, and formation control in, e.g., Cheng, Yue, Hu, Ge, and Chen (2020), Deng and Wen (2020), Feng and Hu (2020), Feng, Hu, and Wen (2016), Senejohnny, Tesi, and De Persis (2017), Tang, Zhang, Shi, Zhang, and Qian (2020) and Xu, Mo, and Xie (2020). In this paper, we will present some of these works for multi-agent systems with different classes of dynamics as well as communication capacities.

We note that in the context of cyber security, it is important to set certain assumptions on the limitations on the capability of the attackers. Such limitations can be determined from past data of attack incidents and will provide the goal for the system operator to prepare security measures. For example, we may want to start with an upper bound on the possible number of malicious agents in the network or the maximum length on the DoS attacks.

1.3. Organization of this paper

This paper is organized as follows. In Section 2, the basics of the problem formulation of resilient consensus as well as the class of distributed protocols known as MSR algorithms are discussed. In Section 3, we provide an introduction to the notion of graph robustness, which plays a key role in resilient consensus. In Section 4, we present the main results characterizing the necessary network requirements for resilient consensus under synchronous and asynchronous communications. Then, Section 5 extends the discussions on the MSR approach to take account of data rate constraints in communication. In Section 6, we proceed further to demonstrate that resilient algorithms can be effective to problem settings involving more dynamics in the agents and adversaries. Section 7 presents some typical applications employing resilient consensus techniques. In Section 8, the discussion focuses on multi-agent consensus under DoS attacks. The paper concludes in Section 9 with an outlook into future challenges.

The notations used for most part of the paper (except Section 8) are Table 1.

2. Resilient consensus: Basic problem settings and algorithms

The first part of this paper is on the problem of resilient consensus for multi-agent systems in the presence of nodes manipulated by adversaries. In this section, we provide the basics of the problem settings including graph related notions, multi-agent consensus and its resilient version, adversary models, and the outline of resilient algorithms.

Table 1

List for notations.

$\mathcal{G}$	Directed graph
$\mathcal{V}$	Set of nodes
$\mathcal{E}$	Set of edges
$\mathcal{N}_i$	Set of incoming neighbors of agent i
$\mathcal{M}_i(k)$	Valid subset of the neighbor set $\mathcal{N}_i$ of agent i
$\mathcal{R}$	Set of regular agents in $\mathcal{V}$
$\mathcal{A}$	Set of adversarial agents in $\mathcal{V}$
n	Number of agents in the graph $\mathcal{G}$
F	Upper bound on the number of adversarial agents
f	Upper bound on the ratio of adversarial agents
$a_{ij}(k)$	Weight between agent i and agent j at time k
h	Time interval within which agents make transmissions
γ	Lower bound for non-zero $a_{ij}(k)$
$x_i(k)$	State value of agent i at time k
$u_i(k)$	Control input of agent i at time k
S	Safety interval $S = [\min, x_i(0), \max, x_i(0)]$
$\mathcal{N}_i^r$	Subset of all nodes having at least r neighbors outside $\mathcal{V}_i \subset \mathcal{V}$
$Q(x)$	Uniform quantizer
T	Sampling period

2.1. Graph notions

In multi-agent systems, the network structure connecting the agents for the communication among them is defined by graphs. Consider the directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consisting of n nodes. Here the set of nodes is denoted by $\mathcal{V} = \{1, 2, \dots, n\}$ and the set of edges by $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$. The edge $(j, i) \in \mathcal{E}$ indicates that node j can send a message to node i , and node j is called an (incoming) neighbor of node i . Let $\mathcal{N}_i = \{j : (j, i) \in \mathcal{E}\}$ be the set of neighbors of node i . The number of neighbors of node i is called its degree and is denoted as $d_i = |\mathcal{N}_i|$.

The path from node i_1 to node i_p is denoted as the sequence of edges $(i_1, i_2, \dots, i_p)$, where $(i_j, i_{j+1}) \in \mathcal{E}$ for $j = 1, 2, \dots, p-1$. The graph $\mathcal{G}$ is said to be strongly connected if for each pair of nodes $i, j \in \mathcal{V}$, there is a path connecting from node i to node j . Moreover, the graph $\mathcal{G}$ is said to have a rooted directed spanning tree if there exists a node from which there are paths to all other nodes of this graph.

The graph $\mathcal{G}$ can be expressed in terms of matrices: Denote by $A_{\mathcal{G}} = [a_{ij}] \in \mathbb{R}^{n \times n}$ its adjacency matrix, whose entries are given by $a_{ij} > 0$ if $(j, i) \in \mathcal{E}$ and $a_{ij} = 0$ otherwise for $i, j \in \mathcal{V}$. For the graph $\mathcal{G}$, its Laplacian matrix $L_{\mathcal{G}} = [l_{ij}] \in \mathbb{R}^{n \times n}$ is given by $l_{ij} = -a_{ij}$ if $i \neq j$ and $l_{ii} = \sum_{j=1}^n a_{ij}$ for $i, j \in \mathcal{V}$. Further, let $\lambda_i, i = 1, 2, \dots, n$, be the eigenvalues of the Laplacian matrix $L_{\mathcal{G}}$.

2.2. Update rules for consensus

For multi-agent systems, the consensus problem forms one of the most basic classes of tasks that can be solved by distributed algorithms. The objective is that the state values held by the agents are exchanged and updated so that all agents eventually reach a common value. While there are various approaches for agents to achieve the goal of consensus, we follow the common class of discrete-time linear algorithms here; see, e.g., Mesbahi and Egerstedt (2010), Olfati-Saber, Fax, and Murray (2007) and Ren, Beard, and Atkins (2007).

Consider the directed graph $\mathcal{G}$ with n nodes. Denote by $x_i(k)$ the value held by agent i at time $k \in \mathbb{Z}_+$. At time k , each agent i collects information from its neighbors in $\mathcal{N}_i$ and updates its own value according to the rule given by

$$x_i(k+1) = x_i(k) + \sum_{j \in \mathcal{N}_i} a_{ij}(k) (x_j(k) - x_i(k)), \quad (1)$$

where the initial values $x_i(0)$ are given and $a_{ij}(k)$ is the possibly time-varying weight for the edge $(j, i) \in \mathcal{E}$. We say that the agents achieve consensus if

$$|x_i(k) - x_j(k)| \rightarrow 0 \text{ as } k \rightarrow \infty \text{ for } i, j \in \mathcal{V}. \quad (2)$$

In the problem setting here, as mentioned above, the weights $a_{ij}(k)$ in the update rule (1) may be time varying. In such a case, to ensure reaching consensus, it is common to make the following assumption.

Assumption 2.1. Regarding the weights $a_{ij}(k)$ in the state update rule in (1), the following three conditions hold:

- (i) There exists a positive integer $h > 0$ such that for each edge $(i, j) \in \mathcal{E}$, the corresponding weight $a_{ij}(k)$ becomes nonzero at least once in the time interval $[\ell h, (\ell + 1)h]$ for $\ell \in \mathbb{Z}_+$.
- (ii) There exists a constant $\gamma > 0$ such that if $a_{ij}(k) \neq 0$, then $a_{ij}(k) \geq \gamma$ for $(j, i) \in \mathcal{E}$ and $k \in \mathbb{Z}_+$.
- (iii) $\sum_{j \in \mathcal{N}_i} a_{ij}(k) < 1$ for $k \geq 0$.

Under these conditions, a necessary and sufficient condition for reaching consensus is that the original digraph $\mathcal{G}$ contains a rooted directed spanning tree (Mesbahi & Egerstedt, 2010; Ren et al., 2007). Note that the conditions (i) and (ii) ensure that the agents interact sufficiently over the network in the long run. There are examples of asymptotically vanishing weights under which consensus is not reached (Lorenz & Lorenz, 2010). Thus, the lower bound γ on the weights in (ii) is needed though its specific value does not have to be known by the agents.

2.3. Adversary models

In the resilient consensus problem, it is assumed that a certain number of agents in the system are adversarial, attempting to prevent consensus among the non-faulty agents. To this end, the agents in $\mathcal{V}$ are partitioned into two sets: $\mathcal{R}$ denotes the set of regular agents and $\mathcal{A} = \mathcal{V} \setminus \mathcal{R}$ represents the set of adversarial agents. The regular agents will follow the designed algorithm properly according to the protocols while the adversarial agents may arbitrarily deviate and make completely different decisions of their own.

More specifically, the adversarial agents can arbitrarily make updates in their state values at any time step k . Here, we take account of the worst case and hence assume that the attacker has access to the global information regarding the entire agent system in real time. Thus, he may exploit the full information of the network topology as well as the update rules and the states of the regular agents. The attacker may also choose any agents as members of $\mathcal{A}$ under some constraints on the number, as specified later. We note that unless explained otherwise, the member agents of $\mathcal{A}$ within the system are fixed throughout the iterations though they are unknown to the regular agents in $\mathcal{R}$. In Section 6.2, we discuss mobile versions of adversaries, which are capable to move from one agent to another over time.

Related to the communication capabilities of the adversaries, there are two models that we should distinguish at this stage. These are the malicious and Byzantine models introduced in, e.g., LeBlanc et al. (2013) and Vaidya et al. (2012) as defined below.

Definition 2.1 (Malicious and Byzantine Agents). An adversarial agent i is said to be *malicious* if it broadcasts its state $x_i(k)$ to all of its neighbors at each transmission. It is said to be *Byzantine* if it is capable to transmit different values as its state $x_i(k)$ to different neighbors at each time k .

Comparing the two adversarial models, we notice that Byzantine agents are more difficult to deal with since they form a larger class being more capable in their transmissions. In fact, the Byzantine model has been more commonly studied in the computer science literature (Dolev et al., 1986; Kieckhafer & Azadmanesh, 1994; Pease et al., 1980). This might be explained since in distributed systems at the time the model was proposed, agents were connected mostly via wired networks.

The malicious model has emerged more recently in the literature and is the focus of this paper. The motivation for considering this weaker model comes, for example, from the applications of wireless

sensor networks (WSNs), where sensor nodes communicate to their neighbors by broadcasting their data. Another relevant example is robotic networks, where the individual robots gain the information of the locations of their peers through the use of vision sensors.

Here, we introduce constraints regarding the adversarial agents in the form of bounds on the numbers of such agents in the network. Specifically, we consider three types of such bounds, which are called the F -total, F -local, and f -fraction local models as defined below.

Definition 2.2 (F -total Model). For $F \in \mathbb{N}$, the set $\mathcal{A}$ of adversarial agents is said to follow the F -total model if $|\mathcal{A}| \leq F$.

Definition 2.3 (F -local Model). For $F \in \mathbb{N}$, the set $\mathcal{A}$ of adversarial agents is said to follow the F -local Model if $|\mathcal{A} \cap \mathcal{N}_i| \leq F$ for each regular agent $i \in \mathcal{V} \setminus \mathcal{A}$.

In this paper, we will mainly discuss the F -total/local models given above, where the number of malicious agents is specified. Another related notion based on fractions can be useful and is defined below.

Definition 2.4 (f -fraction Local Model). For $f \in [0, 1)$, the set $\mathcal{A}$ of adversarial agents is said to follow the f -fraction Local Model if $|\mathcal{A} \cap \mathcal{N}_i| \leq f|\mathcal{N}_i|$ for each regular agent $i \in \mathcal{V} \setminus \mathcal{A}$.

In general, to design security measures, it is critical to make certain assumptions regarding the level of the adversaries. While in practice, it may be difficult to know the exact bound (i.e., the number F or f), one can make estimates from past data and experience and moreover, to be on the sure side, add some safety margins to enhance the safety level.

2.4. General notion of resilient consensus

Now that the model and the scope of the adversarial agents have been clarified, we can formally introduce the notion of resilient consensus for multi-agent systems consisting of both regular and adversarial agents in the network. We start with its simple version, which is stated as an asymptotic property. Later on, we will discuss its different variants.

Definition 2.5 (Resilient Consensus). If for any possible sets and behaviors of the malicious agents and any initial state values of the regular agents, the following conditions are satisfied, then the multi-agent system is said to reach resilient consensus:

1. (*Safety condition*): There exists an interval $S \subset \mathbb{R}$ determined by the initial states $x_i(0)$ of the regular agents such that $x_i(k) \in S$ for all $i \in \mathcal{R}$ and $k \in \mathbb{Z}_+$.
2. (*Consensus condition*): The states of the regular agents asymptotically become equal as

$$\lim_{k \rightarrow \infty} |x_i(k) - x_j(k)| = 0 \text{ for all } i, j \in \mathcal{R}. \quad (3)$$

The resilient version of the consensus problem consists of two important conditions. One is to maintain the regular agents to stay in a safe interval which is known a priori from the initial states of the regular agents. This is imposed to ensure that the behavior of the regular agents remains close to that when no malicious agent is present. In particular, we will show that the interval

$$S = \left[\min_{i \in \mathcal{R}} x_i(0), \max_{i \in \mathcal{R}} x_i(0) \right], \quad (4)$$

containing the initial values of the regular agents is an invariant set for the regular agents. The malicious agents may or may not be in S , but when they stay in S , the regular agents can be influenced by them. The other condition is that under the given adversarial model, the regular agents must asymptotically reach consensus in the presence of misbehaving agents. This is a generalization of the non-resilient case (Mesbahi & Egerstedt, 2010; Olfati-Saber et al., 2007; Ren et al., 2007), where all agents are regular, i.e., $\mathcal{R} = \mathcal{V}$. Usually, only the consensus condition is imposed as the safety condition implicitly holds.

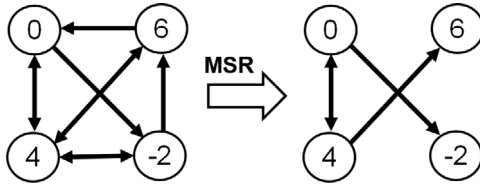


Fig. 1. MSR algorithm with one adversary ($F = 1$).

2.5. Description of MSR algorithms

In the context of multi-agent systems, decisions must be made by each regular agent in a distributed manner. Such an approach is followed in our problem setting of resilient consensus as well. That is, at every time step k , each regular agent i interacts and collects the values from its neighbors and then calculates an updated value $x_i(k+1)$ for itself. A major difference is that a certain number of the neighbors may be adversarial, which is determined by the assumed model. However, the regular agents do not have the means to detect which neighbors may be misbehaving. To ensure that regular agents make updates based on safe information, they take the strategy of selecting values that are not excessively different from their own in the updates. That is, outlying values will be ignored and removed from the data received at each iteration.

For consensus problems with real-valued states, the class of such protocols has been known as the *mean subsequence reduced (MSR)* algorithms. The name of this class has been coined in Kieckhafer and Azadmanesh (1994), which considers agent systems having complete graphs as the underlying networks. Roughly speaking, the MSR algorithm has three main steps, which can be summarized as follows. These steps will appear in different forms throughout this paper. The following algorithm assumes the F -total/local models.

Algorithm 2.1 (MSR Algorithm).

1. (*Interaction with neighbors*): At each time step k , every regular agent $i \in \mathcal{R}$ broadcasts its current state value $x_i(k)$ to its out-neighbors. It then receives the values $x_j(k)$, $j \in \mathcal{N}_i$, communicated from its in-neighbors and sorts them together with its own value $x_i(k)$ from the largest to the smallest.
2. (*Deleting suspicious values*): Comparing with $x_i(k)$, agent i removes the F largest and F smallest values from its neighbors. If the number of values larger or smaller than $x_i(k)$ is less than F , then all of them are removed. The removed data is considered as suspicious and will not be used in the update. The set of the agent indices of the remaining values is written as $\mathcal{M}_i(k) \subset \mathcal{N}_i$.
3. (*Local update*): Agent i updates its state by

$$x_i(k+1) = x_i(k) + \sum_{j \in \mathcal{M}_i(k)} a_{ij}(k) (x_j(k) - x_i(k)). \quad (5)$$

The weight $a_{ij}(k)$ should satisfy the conditions (ii) and (iii) in Assumption 2.1.

As mentioned above, for the adversarial agents, we are assuming the F -total/local malicious models in Algorithm 2.1. Thus, in Step 2 of deletion of suspicious values, each regular agent removes up to F values greater than its current one as well as up to F values smaller than its own. This is a simple operation, but can be effective when there may be F adversaries in the system. In contrast, for the f -fraction local model, in Step 2 above, F should be replaced with $f|\mathcal{N}_i|$, which is the maximum local number of adversarial agents linked to agent i .

Notice that the MSR algorithm does not require any regular agent to have knowledge of the network topology or the capability to detect adversarial agents. In contrast, the regular agents attempt to remove the outlying values among those received from neighbors in real time.

As a result, the set of agents ignored by each regular agent can be time varying, and thus the MSR algorithm induces switching behaviors in the network even though the underlying network topology is fixed. The key problem for MSR algorithms is to establish conditions on the network topologies that guarantee the agents to arrive at resilient consensus. This issue is discussed in the next two sections.

Example 2.1. We present a simple example to illustrate the MSR algorithm using the 4-node graph shown in Fig. 1. The network in the left is the original one and the numbers in the nodes indicate the initial state values of the four agents. In MSR, we take the parameter $F = 1$, i.e., at most 1 malicious agent may be present. After exchanging their values, the agents remove the edges based on MSR, resulting in the network in the right. For example, the agent taking value 4 receives -2 , 0 , and 6 from its neighbors; it will then remove the smallest value -2 and the largest 6 . Hence, after the removal of these values, only one incoming edge from the agent with 0 will remain.

3. Graph robustness

In the characterization of MSR-based algorithms, an important topological notion is that of robustness of graphs (LeBlanc et al., 2013; Zhang, Fata, & Sundaram, 2015; Zhang & Sundaram, 2012b). In this section, we discuss several classes of such graphs as well as their properties, which play important roles to establish the results for resilient consensus. Specifically, the notions of set reachability, r -robustness, (r, s) -robustness, and p -fraction robustness will be introduced from LeBlanc et al. (2013).

3.1. Robust graph definitions

In the context of resilient consensus, it is critical that each agent in the uncertain environment has a sufficient number of peers in the neighborhood so that in any event, the received information is guaranteed to contain some safe data. Such a viewpoint motivates the notion of reachability in graphs as given below.

Definition 3.1 (r -reachability). In the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, given a nonempty subset $\mathcal{V}_1$ of the node set $\mathcal{V}$ and a nonnegative integer $r \in \mathbb{Z}_+$, we say that $\mathcal{V}_1$ is r -reachable if there exists $i \in \mathcal{V}_1$ such that $|\mathcal{N}_i \setminus \mathcal{V}_1| \geq r$.

Clearly, for a node set $\mathcal{V}_1$ to be r -reachable, it is necessary that each node in the set has at least r neighbors. For example, if every node in $\mathcal{G}$ has exactly r neighbors, then the set $\mathcal{V}_1 = \{i\}$ is r -reachable. However, the set $\mathcal{V}_2 = \{i, j\}$ is not if nodes i and j are neighbors of each other.

In our problem setting, to ensure resilience, it is not enough for each node to have a sufficient number of neighbors. This is because the identities of the adversaries are unknown. In fact, we must check every possible pattern on the distribution of adversaries in the network. Such a requirement leads us to a more complex notion called r -robustness for graphs, involving combinatorial aspects using r -reachability.

Definition 3.2 (r -robustness). Given a nonnegative integer $r < n$, the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is called r -robust if for any two nonempty disjoint subsets $\mathcal{V}_1, \mathcal{V}_2 \subset \mathcal{V}$, at least one of the subsets is r -reachable.

Example 3.1. We illustrate the combinatorial nature of graph robustness through an example graph of four nodes in Fig. 2(a). Here, nodes 1, 2, and 4 have two neighbors while node 3 has three. This graph can be checked to be 2-robust. Notice that node sets containing one or two nodes are all 2-reachable. For this property to hold, it is critical that node 3 has three neighbors. Suppose that we remove its incoming edge from node 1, i.e., the edge $(1, 3)$. Then, the node set $\{3, 4\}$ is no longer 2-reachable. However, this graph remains to be 2-robust.

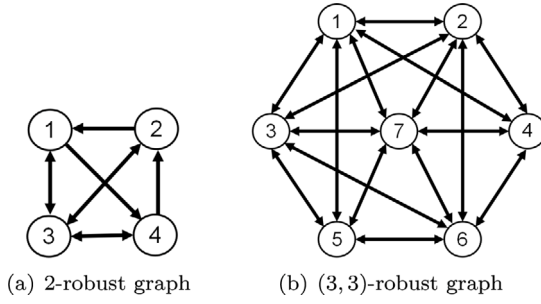


Fig. 2. Robust network topologies.

Finally, we proceed to the notion of (r, s) -robustness, which further extends r -robustness and provides graphs more redundancies. Here, for a given set $\mathcal{V}_1 \subset \mathcal{V}$ of nodes, denote by $\mathcal{N}_{\mathcal{V}_1}^r$ its subset of all nodes having at least r neighbors outside $\mathcal{V}_1$.

Definition 3.3 ((r, s) -robustness). Given nonnegative integers $r, s < n$, the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is called (r, s) -robust if for any two nonempty disjoint subsets $\mathcal{V}_1, \mathcal{V}_2 \subset \mathcal{V}$, one of the following conditions is satisfied:

1. $\mathcal{N}_{\mathcal{V}_1}^r = \mathcal{V}_1$,
2. $\mathcal{N}_{\mathcal{V}_2}^r = \mathcal{V}_2$, and
3. $|\mathcal{N}_{\mathcal{V}_1}^r| + |\mathcal{N}_{\mathcal{V}_2}^r| \geq s$.

In view of the two definitions above, it is clear that r -robustness is a special case of (r, s) -robustness with $s = 1$.

Example 3.2. In Fig. 2(b), we display an example graph with seven nodes. It can be checked to have just enough connectivity to be $(3, 3)$ -robust, and this level of robustness is lost if any edge is removed.

Finally, we present the alternative definitions of robust graphs based on fractions of neighbors for measuring the number of reachable nodes as follows.

Definition 3.4. Given a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ and a nonempty subset $\mathcal{V}_1$ of set $\mathcal{V}$, we say $\mathcal{V}_1$ is p -fraction reachable set if $\exists i \in \mathcal{V}_1$ such that $|\mathcal{N}_i \setminus \mathcal{V}_1| \geq p|\mathcal{N}_i|$, where $0 \leq p \leq 1$.

Definition 3.5. The graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is called p -fraction robust ($0 \leq p \leq 1$) if for any two nonempty disjoint subsets $\mathcal{V}_1, \mathcal{V}_2 \subset \mathcal{V}$, at least one of the subsets is p -fraction reachable set.

3.2. Properties of robust graphs

Here, we discuss some of the properties of robust graphs. The following lemma summarizes the basic results from LeBlanc et al. (2013).

Lemma 3.1. An (r, s) -robust graph $\mathcal{G}$ satisfies the following:

1. $\mathcal{G}$ is (r', s') -robust, where $0 \leq r' \leq r$, $1 \leq s' \leq s$, and in particular, it is r -robust.
2. $\mathcal{G}$ has a directed rooted directed spanning tree. Moreover, it is 1-robust if and only if it has a directed spanning tree.
3. $r \leq \lceil n/2 \rceil$, where the ceiling function $\lceil y \rceil$ gives the smallest integer greater than or equal to y . Furthermore, it holds $r = \lceil n/2 \rceil$ if and only if $\mathcal{G}$ is a complete graph.
4. The degree d_i for $i \in \mathcal{V}$ is lower bounded as $d_i \geq r + s - 1$ if $s < r$ and $d_i \geq 2r - 2$ if $s \geq r$.
5. Suppose $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is an (r, s) -robust subgraph of $\mathcal{G}' = (\mathcal{V}, \mathcal{E}')$, where $\mathcal{E}' = \mathcal{E} \cup \mathcal{E}''$ and $|\mathcal{E}''| \geq 0$. Then $\mathcal{G}'$ is (r, s) -robust.

6. Given an (r, s) -robust (p -fraction robust) graph $\mathcal{G}$, let $\mathcal{G}'$ be the graph produced by removing up to k (q -fraction of) incoming edges of each node in $\mathcal{G}$, where $0 \leq k < r$ ($0 \leq q < p \leq 1$). Then $\mathcal{G}'$ is $(r - k, s)$ -robust ($(p - q)$ -fraction robust).

Moreover, a graph $\mathcal{G}$ is (r, s) -robust if it is $(r + s - 1)$ -robust.

For consensus problems, the second item in Lemma 3.1 above is of interest. Robust graphs may not be strongly connected in general. However, this property indicates that the notion of robust graphs is a generalization of those containing rooted directed spanning trees, which are of great relevance in the literature of consensus (Mesbahi & Egerstedt, 2010).

While the following result from Zhang et al. (2015) is limited to undirected graphs, it shows that robust graphs remain connected even after removing a certain number of nodes. It hence justifies that the notion of robust graphs is critical for securing structural robustness.

Proposition 3.1. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an r -robust undirected graph with $r \geq 1$. Let $\mathcal{V}_1 \subset \mathcal{V}$ be a subset with the property $|\mathcal{N}_i \cap \mathcal{V}_1| \leq r - 1$ for all i . Further, let $\mathcal{G}' = (\mathcal{V} \setminus \mathcal{V}_1, \mathcal{E}')$ be the subgraph of $\mathcal{G}$ after removing the nodes in $\mathcal{V}_1$ and their incident edges. Then, $\mathcal{G}'$ is connected.

When a given graph does not exhibit a sufficient level of robustness, there are different approaches to enhance it. We outline three options in the following. Notice that they all incur additional costs for realizing further resilience in the system.

1. A conceptually simple solution is to establish new edges among agents, which enables more communication. Further, deploying additional agents can be useful as they create more paths in the system for safe transmissions of information. For related discussions, see Section 3.3.
2. The second approach is to introduce the so-called *trusted agents* (Abbas, Laszka, & Koutsoukos, 2017). The main idea is to ensure that a small subset of agents, referred to as the trusted agents, remains intact not turning malicious or Byzantine and functions correctly at all times. By controlling the number and location of trusted agents, any desired connectivity and robustness can be achieved without adding extra edges.
3. Yet another approach is to allow the agents to employ *multi-hop communication* (Goldsmith, 2005). Agents will then receive data not only from their direct neighbors but also from multi-hop neighbors through relaying of messages. It has been used in the context of multi-agent consensus (Jin & Murray, 2006; Manfredi, 2013). In resilient consensus problems, though adversaries may be able to falsify the data to be relayed, robustness can generally be increased as discussed in Su and Vaidya (2017) and Yuan and Ishii (2021a, 2021b).

3.3. Verification and construction of robust graphs

In this subsection, we briefly provide discussions related to the analysis and design of robust graphs, for which there is a line of recent research.

A critical problem for robust graphs is to find the level of robustness for a given graph $\mathcal{G}$. From item 3 of Lemma 3.1, an obvious upper bound on its r -robustness can be found to be $r \leq \lceil n/2 \rceil$, where n is the number of agents in $\mathcal{G}$. However, as seen from the previous subsections, this problem in general can be computationally expensive. This is because in the definitions of r - and (r, s) -robustness, we must check the reachability of all pairs of agent subsets in the graph. In fact, it is known that determining the robustness of an arbitrary network is an NP-hard problem (LeBlanc & Koutsoukos, 2013; Zhang & Sundaram, 2012b). To deal with the computational issues involved, the work by Usevitch and Panagou (2020) developed mixed integer programming methods, which enable us to find bounds on the parameters r and s for a graph at specified error levels.

Moreover, the problem of finding robustness for some special classes of graphs has been approached. For example, [Usevitch and Panagou \(2017\)](#) have focused on circulant graphs and proposed methods to check their robustness. Different random graphs have been studied as well. The works ([Zhang et al., 2015](#); [Zhang & Sundaram, 2012a](#)) characterized graph robustness of random graphs such as Erdős–Rényi, 1-D geometric, and Barabási–Albert random graphs. Other works carried out analyses on random intersection graphs ([Zhao, Yağan, & Gligor, 2016](#)) and random interdependent networks ([Shahrivar, Pirani, & Sundaram, 2017](#)).

On the other hand, the problem of designing robust graphs having the desired robustness property is of interest especially from the engineering viewpoint. For example, in [Saldana, Prorok, Campos, and Kumar \(2018\)](#), a method for constructing triangular graphs of arbitrary sizes that are $(2, 2)$ -robust is introduced. In [LeBlanc et al. \(2013\)](#) and [Zhang and Sundaram \(2012b\)](#), the problem is approached based on that the common preferential-attachment model for complex networks. That is, we start from a small-sized graph having the desired level of robustness and then attach new agents to the graph one by one along with necessary numbers of edges. The work ([Saldana et al., 2018](#)) derived lower bounds on the number edges for producing r -robust and (r, s) -robust graphs as summarized below.

Theorem 3.1. *Let $G = (\mathcal{V}, \mathcal{E})$ be an r -robust graph. Then, the graph $G' = (\{\mathcal{V}, v'\}, \{\mathcal{E}, \mathcal{E}'\})$, where v' is a new vertex added to G and $\mathcal{E}'$ is the edge set related to v' , is r -robust if $d_{v'} \geq r$.*

Theorem 3.2. *Let $G = (\mathcal{V}, \mathcal{E})$ be an (r, s) -robust graph. Then, the graph $G' = (\{\mathcal{V}, v'\}, \{\mathcal{E}, \mathcal{E}'\})$ where v' is a new vertex added to G and $\mathcal{E}'$ is the edge set related to v' , is (r, s) -robust if $d_{v'} \geq r + s - 1$.*

4. Fundamental results for MSR algorithms

In this section, we would like to highlight the fundamental results for MSR algorithms under several settings in terms of the agents' protocols and the adversarial models.

Here, we deal with the synchronous case, where all agents make transmissions and updates at each discrete-time step. The results to be presented characterize the necessary connectivity of the underlying networks over which the agents interact to guarantee resilient consensus. In particular, the conditions are represented by the notion of robust graphs, introduced in the previous section. For all of these results, the safety interval S can be taken as the one in (4).

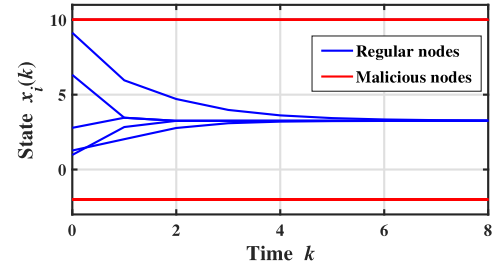
4.1. Synchronous protocols under malicious adversaries

Here, we focus on the malicious model for the adversaries. In what follows, results for three different adversary models are presented from [LeBlanc et al. \(2013\)](#).

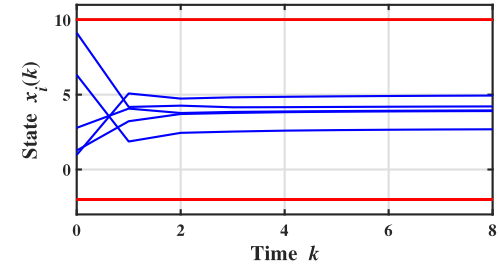
The most fundamental is the result for the F -total malicious model, in which case at most F adversaries may be part of the entire network. For this model, a tight necessary and sufficient condition on the network connectivity can be attained as shown in the following theorem.

Theorem 4.1. *Under the F -total malicious model, the regular agents with the MSR algorithm reach resilient asymptotic consensus if and only if the underlying graph is $(F + 1, F + 1)$ -robust.*

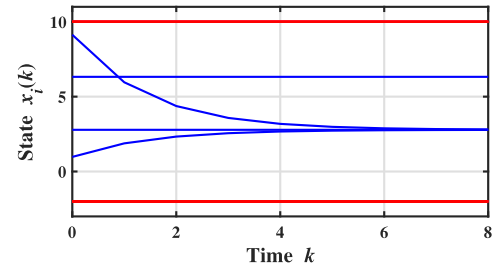
The next result is applicable to the F -local version of the malicious model. More adversaries are allowed in the system than the F -total case since each regular agent may have up to F adversaries among its neighbors. As a consequence, the network conditions become more restrictive in comparison to those for the F -total model.



(a) MSR algorithm with $F = 2$ under 2 malicious agents



(b) Conventional algorithm under 2 malicious agents



(c) MSR algorithm with $F = 3$ under 3 malicious agents

Fig. 3. Time responses of all agents in the network.

Theorem 4.2. *Under the F -local malicious model, the regular agents with the MSR algorithm reach resilient asymptotic consensus if the underlying graph is $(2F + 1)$ -robust and only if the underlying graph is $(F + 1, F + 1)$ -robust.*

It is noted that $(2F + 1)$ -robust graphs are also $(F + 1, F + 1)$ -robust by [Lemma 3.1](#). This indicates the presence of a gap between [Theorems 4.1](#) and [4.2](#) for the F -total model scheme and the F -local model, respectively. This gap is in fact known to be nonconservative and counter examples have been presented in [LeBlanc et al. \(2013\)](#).

The third case provides an alternative view on the problem as it deals with the f -fraction local malicious model. Recall that in this case, for each regular agent, at most the f fraction of its neighbors may be adversarial. In the following theorem, the network conditions are expressed in terms of the corresponding notion of p -fraction robust graphs.

Theorem 4.3. *Under the f -fraction local malicious model, the regular agents with the MSR algorithm reach resilient asymptotic consensus if the underlying graph is p -fraction robust, where $2f < p \leq 1$ and only if the underlying graph is p' -fraction robust, where $p' > f$.*

Example 4.1. We present a simple example using the 7-node network in [Fig. 2\(b\)](#). As mentioned earlier, this network is $(3, 3)$ -robust. Hence, according to [Theorem 4.1](#), the MSR algorithm applied to this system can tolerate up to 2 malicious agents by using $F = 2$ in the protocol.

First, to confirm this property, we set agents 1 and 2 to be malicious, taking constant state values 10 and -2 , respectively. Other agents take initial values in the interval $[0, 10]$.

The time responses of the states from the MSR algorithm are shown in Fig. 3(b). We see that the agents reach consensus despite the constant values taken by some neighbors. Second, Fig. 3(a) shows results when the conventional consensus algorithm is used (i.e., with $F = 0$ in the MSR). Here, the agents come close in their values, but errors among regular agents remain. Finally, we turn another agent malicious and apply the MSR with $F = 3$. By letting agent 3 take the constant value of 10, the results displayed in Fig. 3(c) indicate that the agents are divided into two groups, settling at values around 3 and 6. It is clear that the network in this case has insufficient connectivity for the MSR with $F = 3$.

It is interesting to note that two agents in the network have in-degree of 4, which means that using the MSR with $F = 2$, they may ignore all neighbor values. Also, as mentioned earlier, the network will lose the property of being $(3, 3)$ -robust if any edge is removed; in such a case, even with 2 malicious agents, the division in the values of the regular agents will occur, similarly to the dynamics shown in Fig. 3(c).

4.2. Asynchronous protocols under malicious adversaries

We next turn our attention to an asynchronous version of the MSR algorithm. In this setting, the agents interact with each other and make updates at different time instants.

An asynchronous approach common in the computer science literature (e.g., Kieckhafer & Azadmanesh, 1994; Vaidya et al., 2012), which is also employed in LeBlanc and Koutsoukos (2012), is based on the technique of rounds. There, each agent broadcasts its state and its round, which is the number of transmissions that the agent made so far. The agent makes an update at round $r + 1$ by using the states of neighbors, but only when a sufficient number of data from round r is received. Note that states from round r might be received at different times, which can cause potentially large time delays.

To this end, the modifications consist of the following two points: (i) The transmitted messages are tagged with the corresponding round index, and (ii) each regular agent j waits to receive only $d_j^* = d_j - F$ messages from in-neighbors for a given round r before updating its value. The following two theorems give sufficient conditions in terms of robust digraphs to guarantee resilient asymptotic consensus with asynchronous MSR algorithms.

The first result is for the F -total malicious model.

Theorem 4.4. *Under the F -total malicious model, the regular agents with asynchronous MSR reach resilient asymptotic consensus if the underlying graph is $(2F + 1, F + 1)$ -robust.*

It is clear that more robustness is needed to safely run the asynchronous algorithm. The second one is for the case of F -local malicious model, which calls for even further connectivity.

Theorem 4.5. *Under the F -local malicious model, the regular agents with the asynchronous MSR reach resilient asymptotic consensus if the underlying graph is $(3F + 1)$ -robust.*

We should emphasize that the asynchronous protocols based on rounds are inefficient not only in their convergence speeds due to the delays mentioned above, but also in the network conditions stated in the theorems. In this paper, several approaches will be discussed for asynchronous interactions among agents without relying on the mechanism of rounds. For example, Section 5.1 presents a protocol using event-triggered control to reduce the frequency of communication. Moreover, we will see in Section 5.2 that randomization-based protocols have significant advantages regarding the necessary network connectivities.

4.3. Byzantine adversaries

Next, we switch the class of adversaries to that of the Byzantine type, where the adversarial agents are capable to transmit messages containing different data to different neighbors at the same time.

Here, we present a brief overview of Vaidya et al. (2012), which has investigated the usefulness of the MSR algorithms in the context of F -total Byzantine adversaries. In particular, a necessary and sufficient condition on the graph structure for the algorithm to succeed can be obtained. Clearly, in comparison to the malicious model, the attack level is higher. We will see that a more dense connectivity in the network is needed to protect the system from adversaries of Byzantine model.

Before we proceed, another graph notion is introduced, which is related to the subgraph consisting of only regular agents.

Definition 4.1. For a network $\mathcal{G}$, define the regular network of $\mathcal{G}$ denoted by $\mathcal{G}_R$ as the network induced by the regular agents, i.e., $\mathcal{G}_R = (\mathcal{R}, \mathcal{E}_R)$, where $\mathcal{E}_R$ is the set of directed edges among the regular agents.

For the F -total model, a tight condition on the network structure of regular agents is attained as shown in the following result.

Theorem 4.6. *Under the F -total Byzantine model, the regular agents with the MSR algorithm reach resilient asymptotic consensus if and only if the topology of the regular network is $(F + 1)$ -robust.*

Based on this theorem, it is relatively simple to obtain an extension for the F -local model.

Theorem 4.7. *Under the F -local Byzantine model, the regular agents with the MSR algorithm reach resilient asymptotic consensus if and only if the topology of the regular network is $(F + 1)$ -robust.*

Finally, the counterpart for the f -fraction model is shown below.

Theorem 4.8. *Under the f -fraction local Byzantine model, resilient asymptotic consensus is achieved by the MSR algorithm if the regular network is p -fraction robust, where $p > f$, and a necessary condition is for the regular network to be p' -fraction robust, where $p' \geq f$.*

5. Resilient consensus with resource limitations

In this section, we present several approaches for achieving resilient consensus when resources are limited for the agents in terms of computation and communication.

5.1. Event-triggered resilient consensus

To reduce the frequency of transmissions by the agents in a network, event-triggered protocols are known to be effective and have been studied extensively (Dimarogonas, Frazzoli, & Johansson, 2012; Guo, Ding, & Han, 2014; Kadowaki & Ishii, 2014; Meng & Chen, 2013; Seyboth, Dimarogonas, & Johansson, 2013). In this subsection, we introduce an event-based protocol for the regular agents to achieve consensus in the presence of malicious agents. The algorithm is called the *event-based mean subsequence reduced (E-MSR) algorithm* (Wang & Ishii, 2019b). It can be seen as one of the asynchronous versions of Algorithm 2.1 in Section 2.5.

The E-MSR algorithm can be outlined as follows: At each time step $k \in \mathbb{Z}_+$, the agents make updates in a synchronous manner, but whether they transmit their current values to neighbors or not depends on their triggering functions. More concretely, each agent i has an auxiliary variable, which is its state value communicated the last time, and compares it with its own state of current time. If the current state has changed sufficiently and the difference from the past value is large, then the agent will send it to its neighbors and also replace the auxiliary variable with it.

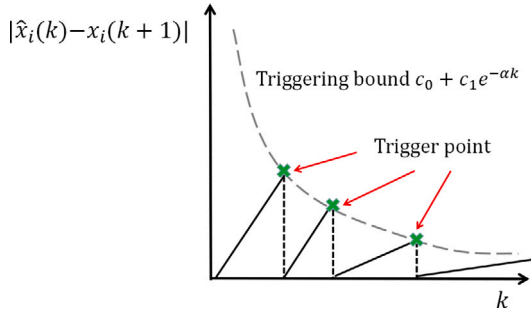


Fig. 4. Event-triggered communication.

To accommodate the event-triggered communication, some of the steps in the MSR algorithm (Algorithm 2.1) will be modified. For those steps, we need some new notation: First, let $t_0^i, t_1^i, \dots$ denote the transmission times of agent i , which are determined by the triggering rule to be given below. Next, define the auxiliary state $\hat{x}_i(k) \in \mathbb{R}$ representing the last communicated state of agent i at time k :

$$\hat{x}_i(k) = x_i(t_l^i), \quad k \in [t_l^i, t_{l+1}^i).$$

Then, we introduce the triggering function for agent i :

$$f_i(k) = |\hat{x}_i(k) - x_i(k+1)| - (c_0 + c_1 e^{-\alpha k}), \quad (6)$$

where c_0, c_1 , and $\alpha > 0$ are positive constants. As we will see below, events will be triggered at time instants when this function becomes positive. These are moments when the error between the new state $x_i(k+1)$ and the last communicated state $\hat{x}_i(k)$ exceeds the threshold $c_0 + c_1 e^{-\alpha k}$ (see Fig. 4). The update rule in (5) needs to be slightly modified as well.

For the E-MSR algorithm, the following show the new Steps 3 and 4 to be used in Algorithm 2.1:

3'. (Local update): Agent i updates its state by

$$x_i(k+1) = x_i(k) \sum_{j \in \mathcal{M}_i(k)} a_{ij}(k) (\hat{x}_j(k) - x_i(k)). \quad (7)$$

4'. (Transmission at event times): If $f_i(k) > 0$, then agent i transmits its new state $x_i(k+1)$ to the neighbor agents at time k . Moreover, it sets the transmission time as $t_l^i = k$, where $t_{l-1}^i < k$, and the auxiliary variable as $\hat{x}_i(k) = x_i(k+1)$.

We discuss a few points regarding the triggering function in (6). Notice that because of the exponentially decreasing term $c_1 e^{-\alpha k}$, the time interval between the events may be large at the early stage during the consensus process, but then decreases as time passes by. On the other hand, due to the constant term c_0 in (6), the time intervals between events are always lower bounded while the difference between $\hat{x}_i(k)$ and $x_i(k+1)$ will persist, and thus, it is hard to reach perfect consensus as in (3) of Definition 2.5.

Hence, we must relax the goal to resilient consensus with a given error level $c > 0$. In particular, the consensus condition in Definition 2.5 is replaced with the following while the safety condition remains the same:

2'. (Consensus condition with error level c): The error between the states of the regular agents eventually become smaller than $c > 0$ as

$$\limsup_{k \rightarrow \infty} |x_i(k) - x_j(k)| \leq c \text{ for all } i, j \in \mathcal{R}. \quad (8)$$

The specific problem to be addressed in the design of the E-MSR algorithm is as follows: Along with the topological condition on the network connectivity, find the parameter c_0 in the triggering function in (6) for achieving resilient consensus with the error level c as in (8).

This problem is fully solved in the next theorem.

Theorem 5.1. Under the F -total malicious model, the regular agents with the E-MSR algorithm reach resilient consensus at an error level c if and only if the underlying graph is $(F+1, F+1)$ -robust. The safety interval is given by $S = [\min_{i \in \mathcal{R}} \hat{x}_i(0), \max_{i \in \mathcal{R}} \hat{x}_i(0)]$, and the consensus error level c is achieved if the parameter c_0 in the triggering function (6) satisfies

$$c_0 \leq \frac{\gamma^N c}{4N}. \quad (9)$$

The communication rule in this algorithm shows that only when the current value has varied enough to exceed a threshold, then the auxiliary variable will be updated, and only at this time the agent sends its value to its neighbors. This event triggering scheme can significantly reduce the communication burden with limited effects on the convergence rate.

It is interesting to point out that the E-MSR algorithm can be slightly modified to enhance its performance. Notice that in the update rule (7) for obtaining the new state $x_i(k+1)$ of agent i , its own data appears only through the current state $x_i(k)$. On the one hand, this means that even when the new state is not communicated, it still needs to be stored at every time step. On the other hand, as the current state $x_i(k)$ is newer than $\hat{x}_i(k)$, it seems desirable for speeding up the convergence. However, it has been shown that it may be advantageous to use only $\hat{x}_i(k)$ for both storage and convergence reasons; for further details about such an approach in consensus problems, we refer the readers to, e.g., Kadowaki and Ishii (2014) and Wang and Ishii (2019b).

5.2. Quantized resilient consensus: Randomized approach

In this subsection, we introduce the network of agents whose states take integer (i.e., quantized) values under directed communication links. Quantized consensus has been motivated by concerns on limited capabilities in communications and computations of the agents. For recent works on this subject, see, for example, Aysal, Coates, and Rabbat (2008), Cai and Ishii (2011), Carli, Fagnani, Frasca, and Zampieri (2010), Kashyap, Başar, and Srikant (2007), Lavaei and Murray (2012) and Nedec, Olshevsky, Ozdaglar, and Tsitsiklis (2009). Resilient versions have been developed in Dibaji et al. (2018), where asynchronous protocols without the use of rounds can be found. The work (Wang & Ishii, 2020) further studies the event-based protocol for the quantized consensus under adversarial attacks. In what follows, we present the results there.

We start with the synchronous protocol. The problem setting is mostly the same as that for the real-valued state case discussed in Section 2. Here, all agents take states $x_i(k) \in \mathbb{Z}$ of integer values at each time step k . This requires us to introduce the quantization function $Q: \mathbb{R} \rightarrow \mathbb{Z}$. In particular, we employ a probabilistic uniform quantizer (Aysal et al., 2008) defined as follows:

$$Q(x) = \begin{cases} \lfloor x \rfloor & \text{with probability } p(x), \\ \lceil x \rceil & \text{with probability } 1 - p(x), \end{cases} \quad (10)$$

where $p(x) = \lceil x \rceil - x$. For quantized consensus, it is known that deterministic quantizers and protocols may lead the agent systems to problematic situations where consensus cannot be attained (Cai & Ishii, 2011; Kashyap et al., 2007).

For the quantized MSR algorithm, the update step in Algorithm 2.1 must be changed to the following one using the quantizer:

3''. (Local update): Agent i updates its state by

$$x_i(k+1) = x_i(k) + Q\left(\sum_{j \in \mathcal{M}_i(k)} a_{ij}(k) (x_j(k) - x_i(k))\right). \quad (11)$$

We now discuss the notion of resilient consensus for the network of quantized agents in the presence of misbehaving agents. While due to the use of the probabilistic quantizer, the notion of consensus must be probabilistic as well, consensus forming can be completed in finite time. Specifically, for the quantized case, we replace the consensus condition in Definition 2.5 with the following one, where the safety condition remains the same:

2''. (Quantized consensus condition): There exists a finite time $k_a \geq 0$ such that

$$\text{Prob}\{x_i(k_a) = x_j(k_a), \forall i, j \in \mathcal{R} \mid x(0)\} = 1.$$

Then, the main result for the synchronous update case can be stated as follows [Dibaji et al. \(2018\)](#). Here, the safety interval S is given by (4) similarly as in the previous results.

Theorem 5.2. *Under the F -total malicious model, the network of quantized agents with the quantized MSR algorithm reaches resilient consensus almost surely with respect to the randomized quantization if and only if the underlying graph is $(F + 1, F + 1)$ -robust.*

This theorem presents a tight necessary and sufficient condition on the network topology. The network condition remains the same as those in the results such as [Theorems 4.1](#) and [5.1](#) that we have seen so far for the real-valued protocols though resilient consensus is now guaranteed in a probabilistic sense. While this may seem natural, we will see next that this nature can change once we consider asynchronous schemes and compare deterministic and probabilistic protocols.

As asynchronous protocols, we introduce two types which differ in the updating timings for the agents. The first one is deterministic, where each regular agent i makes an update at least once in T time steps. Recall the first condition in [Assumption 2.1](#). This condition can be alternatively expressed as

$$\bigcup_{l=k}^{k+h-1} \mathcal{U}(l) = \mathcal{V} \setminus \mathcal{A}, \forall k \in \mathbb{Z}_+, \quad (12)$$

where $\mathcal{U}(k)$ is the set of regular agents which make updates at time k .

When the MSR algorithm in [Algorithm 2.1](#) follows this deterministic updating scheme, we arrive at the result given below.

Theorem 5.3. *Under the F -total malicious model, the network of asynchronous quantized agents with the quantized MSR algorithm satisfying (12) reaches resilient quantized consensus almost surely with respect to the randomized quantization if the underlying graph is $(2F + 1)$ -robust.*

Observe that under the deterministic asynchronous case, the network condition has become more restrictive. In comparison with the synchronous case in [Theorem 5.2](#) for quantized consensus, the condition changed from $(F + 1, F + 1)$ -robust to $(2F + 1)$ -robust. We must point out that this latter condition is still more relaxed than the one for the asynchronous protocol using the round scheme presented in [Theorem 4.4](#); the condition there was $(2F + 1, F + 1)$ -robust. The difference in the state values between real and quantized plays no role for this issue (see also [Dibaji & Ishii, 2015a](#)).

We must highlight that the condition on network connectivities can be further relaxed by a protocol based on probabilistic updates. In fact, as we see next, the network condition of $(2F + 1)$ -robustness can be replaced with the minimal one, i.e., $(F + 1, F + 1)$ -robustness, which coincides with that for the synchronous case.

Under the probabilistic update rule, each regular agent i makes an update at time $k \geq 0$ with probability $p_i \in (0, 1]$ in an independent and identically distributed (i.i.d.) fashion. More specifically, for agent i , at each time k

$$\text{Prob}\{i \in \mathcal{U}(k)\} = p_i, \quad \text{Prob}\{i \notin \mathcal{U}(k)\} = 1 - p_i. \quad (13)$$

It is noted that the algorithm based on such updates is sometimes referred to as *gossiping* ([Tempo & Ishii, 2007](#)) and can be realized in a fully distributed fashion. Furthermore, the probabilities p_i need not be identical among the agents.

In the context of cyber security, randomization in updating times can be seen as a protective measure for the network. In fact, it makes it difficult for the adversaries to predict the behavior of the regular agents in terms of their transmission instants. Moreover, there is always

nonzero probability that all regular agents within the network simultaneously make updates. This property allows us to establish a stronger result than that for the deterministic case.

The theorem below is the main result of this subsection. It shows that for the probabilistic updating scheme, the condition on graph robustness is the same as that for the synchronous case, recovering the minimal requirement in [Theorem 5.2](#).

Theorem 5.4. *Under the F -total malicious model, the network of asynchronous quantized agents with the quantized MSR algorithm satisfying (13) reaches resilient quantized consensus almost surely with respect to the randomized quantization if and only if the underlying graph is $(F + 1, F + 1)$ -robust.*

The results presented so far show that randomization is crucial in resilient consensus. In computer science, randomization has been used in various algorithms to improve efficiency ([Motwani & Raghavan, 1995](#)) and has been well explored in problems of binary-valued consensus under adversarial attacks ([Lynch, 1996](#)); see also [Tempo, Calafiore, and Dabbene \(2013\)](#) and [Tempo and Ishii \(2007\)](#) for randomized algorithms applied to control problems. For example, in the so-called Byzantine general problem, randomization is necessary to obtain algorithms scalable in convergence times with respect to the size of the networks ([Motwani & Raghavan, 1995](#); [Rabin, 1983](#)). It is noted that the quantized algorithm discussed in this subsection can be applied to the binary-valued case by limiting the initial states to 0 and 1.

For the cases that the agents have access to only delayed information of the neighbors' states, resilient quantized consensus can be achieved in both synchronous and asynchronous networks ([Dibaji et al., 2018](#)). Time delays in communication have certain effects on the requirement for the underlying networks. In fact, it coincides with the condition for the asynchronous protocol with deterministic updates in [Theorem 5.3](#), that is, $(2F + 1)$ -robust graphs. This condition remains the same regardless of the update schemes, meaning that randomization does not help. Moreover, [Wang and Ishii \(2020\)](#) introduced the event-based protocols to achieve resilient quantized consensus. The relationship between the parameters in the triggering function and the consensus error level is investigated.

6. Resilience in consensus for agents with further dynamics and distributed optimization

In this section, we present different classes of multi-agent problems where MSR-type algorithms can be effective in protecting the systems from malicious adversaries. These problems involve more dynamics in the agents, adversaries, or the update laws of the agents. We also address problems where the agents collectively reach optimal states. Namely, we discuss the following three cases: (i) agents with second-order dynamics, (ii) adversaries capable to change the agents to be attacked, and (iii) distributed optimization under attacks.

6.1. Second-order resilient consensus

In this subsection, we introduce a resilient consensus approach for a network of agents having second-order dynamics from [Dibaji and Ishii \(2015b, 2017\)](#). This agent model exhibits more complicated behavior in comparison to the single-order case, making the analysis more challenging. Autonomous mobile robots and vehicles are typical examples of second-order systems ([Francis & Maggiore, 2007](#); [Ren & Cao, 2011](#)). These applications provide motivations different from those in computer science.

Consider a network of agents whose interactions are represented by the directed graph $\mathcal{G}$. Each agent $i \in \mathcal{V}$ has a double-integrator dynamics given by

$$\dot{x}_i(t) = v_i(t), \dot{v}_i(t) = u_i(t), i \in \mathcal{V}, \quad (14)$$

where $x_i(t), v_i(t) \in \mathbb{R}$ are its position and velocity, respectively, and $u_i(t)$ is the control input. We discretize the system with sampling period $T > 0$ as

$$\begin{aligned} x_i(k+1) &= x_i(k) + Tv_i(k) + \frac{T^2}{2}u_i(k), \\ v_i(k+1) &= v_i(k) + Tu_i(k), i \in \mathcal{V}, \end{aligned} \quad (15)$$

where $x_i(k), v_i(k), u_i(k) \in \mathbb{R}$ are, respectively, the position, the velocity, and the control input of agent i at $t = kT$ for $k \in \mathbb{Z}_+$. The discretization is based on control inputs generated by the zeroth order holds.

At each time step k , the regular agents update their positions and velocities based on the time-varying topology of the graph $\mathcal{G}(k)$, which is a subgraph of $\mathcal{G}$ and is specified later. In particular, the control uses the relative positions with its neighbors and its own velocity:

$$u_i(k) = \sum_{j \in \mathcal{N}_i(k)} a_{ij}(k) [(x_i(k) - \delta_i) - (x_j(k) - \delta_j)] - \alpha v_i(k), \quad (16)$$

where α is a positive scalar, and $\delta_i \in \mathbb{R}$ is a constant representing the desired relative position of agent i in a formation (Dibaji & Ishii, 2015b, 2017; Ren & Cao, 2011).

The objective of the regular agents is to reach consensus in the sense that they come to formation expressed in the parameters $\{\delta_i : i \in \mathcal{V}\}$ and then stop asymptotically:

$$x_i(k) - x_j(k) \rightarrow \delta_i - \delta_j, \quad v_i(k) \rightarrow 0 \text{ as } k \rightarrow \infty, \forall i, j \in \mathcal{V}.$$

The notion of resilient consensus for the network of second-order agents is based on this objective as formally stated below. Here, introduce the new state in terms of the error from the desired position as $\hat{x}_i(k) = x_i(k) - \delta_i$.

Definition 6.1. If for any possible set of malicious agents, any initial positions and velocities, and any malicious inputs, the following conditions are met, then the network is said to reach *second-order resilient consensus*:

1. (*Safety condition on positions*): There is an interval $S \subset \mathbb{R}$ such that $\hat{x}_i(k) \in S$ for all $i \in \mathcal{R}$, $k \in \mathbb{Z}_+$.
2. (*Second-order consensus condition*): For some $c \in S$, it holds that $\lim_{k \rightarrow \infty} \hat{x}_i(k) = c$ and $\lim_{k \rightarrow \infty} v_i(k) = 0$ for all $i \in \mathcal{R}$.

We remark that the safety interval is imposed only for the positions of the regular agent and not for their velocities. This indicates that they may move faster than their initial speeds during the formation process. On the other hand, the requirement for the agents to stop at the end is important; consensus in velocity at a safe value is difficult to realize in the presence of adversaries.

The algorithm proposed in Dibaji and Ishii (2015b, 2017) to reach second-order resilient consensus is called *DP-MSR*, which stands for *double-integrator position-based mean subsequence reduced* algorithm. A few changes are necessary in the three steps of the MSR of Algorithm 2.1 as follows:

- (i) In the first interaction step, at time k , each regular agent i receives the relative position values $\hat{x}_j(k) - \hat{x}_i(k)$ of its neighbors $j \in \mathcal{N}_i$ and sorts them in a decreasing order.
- (ii) In the deletion step, each regular agent i focuses on the relative position values $\hat{x}_j(k) - \hat{x}_i(k)$ of the neighbors $j \in \mathcal{N}_i$ and remove the F largest and the F smallest among them.
- (iii) In the local update step, regular agent i applies the control in (16) to its own dynamics in (15).

For the choices of the sampling period T and the parameter α in the control input (16), they must satisfy the following relations:

$$1 + \frac{T^2}{2} \leq \alpha T \leq 2 - \frac{T^2}{2}.$$

This condition is introduced to avoid undesired oscillatory behaviors in the agent movements, and similar conditions appear in, e.g., Qin and Gao (2012) and Ren and Cao (2011).

The safety interval S is given by

$$S = \left[\min_{i \in \mathcal{R}} \hat{x}_i(0) + \min_{i \in \mathcal{R}} \left\{ 0, \left(T - \frac{\alpha T^2}{2} \right) v_i(0) \right\}, \max_{i \in \mathcal{R}} \hat{x}_i(0) + \max_{i \in \mathcal{R}} \left\{ 0, \left(T - \frac{\alpha T^2}{2} \right) v_i(0) \right\} \right]. \quad (17)$$

Note that, as in the previous cases, the interval is determined only by the initial states of the regular agents, but here it involves both their positions and velocities.

Now, we are ready to state the main result for the second-order agent network.

Theorem 6.1. Under the F -total malicious model, the network of agents with second-order dynamics using the DP-MSR algorithm reaches second-order resilient consensus if and only if the underlying graph is $(F + 1, F + 1)$ -robust. The safety interval is given by (17).

Similarly, a result for the F -local case can be derived. The condition on the underlying network of the agents for this case is $(2F + 1)$ -robustness. As discussed in Dibaji and Ishii (2017), the F -local model is more difficult to analyze partly because there are more adversaries in the system in comparison with the F -total model. In the reference above, further studies are performed for networks with partial asynchrony and delays in communication.

6.2. Malicious attacks with mobility

So far, we have focused on multi-agent consensus problems in a hostile environment under the worst-case attacks launched by adversaries. Indeed, the adversaries are fairly powerful having the global information regarding the network topology and the agent's data in real time. In this subsection, we consider adversaries having further capabilities to move within the system and change the agents with faulty behaviors though the total number of such adversaries remain bounded. The material is taken from the work (Wang, Ishii, Bonnet, & D  fago, 2022b).

Specifically, we study adversaries which are able to switch the target agents from time to time. Such mobile adversaries can communicate and collaborate with each other even without any direct links connecting them in the original network. An interesting feature here is that when the adversary leaves, the attacked agent may recover and become fault-free again. However, at the time of recovery, the state of such an agent might remain contaminated. Such a recovered agent will take actions at different levels depending on its awareness of the recovery. One approach would be to restart and join the consensus process by using only the current states of the neighbors. This recovery action may be done through reboot or reset of the system, which can be performed by the system operator or by devices such as watchdogs (Amin et al., 2009).

For mobile adversaries, several models have been proposed in the computer science literature (Bonnet, D  fago, Nguyen, & Potop-Butucaru, 2016; Buhrman, Garay, & Hoepman, 1995; Garay, 1994; Sasaki, Yamauchi, Kijima, & Yamashita, 2013). Recent studies dealing with agents of real-valued states include (Bonomi, Del Pozzo, Potop-Butucaru, & Tixeuil, 2019; Sakavalas & Tseng, 2018; Tseng, 2017). From our perspective, all of these works have common limitations: The underlying networks are assumed to be complete graphs and, moreover, only Byzantine type adversaries are considered.

In what follows, we discuss the problem for the class of malicious adversaries over general directed graphs. To this end, we introduce a new class for such faulty agents, which is called the *mobile malicious model*. Informally, this class exhibits the following three features:

1. Adversarial agents may transmit their false states to their neighbors through broadcasting, that is, all neighbors of a malicious agent receive the same data from it.

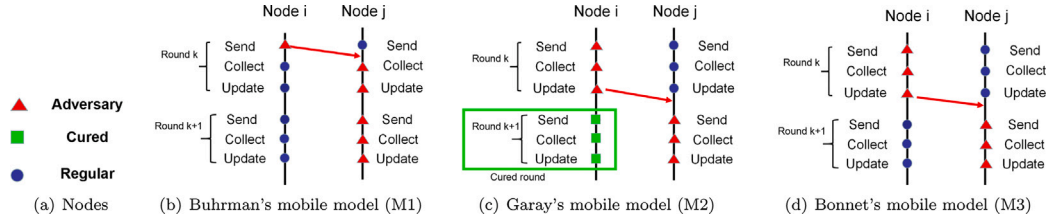


Fig. 5. Mobile adversary models for the malicious agents case.

Table 2

Comparison of mobile malicious/Byzantine models and sufficient conditions for real-valued resilient consensus.

Model of adversaries	Timing of infection	Awareness of becoming cured	Network condition			
			Complete graphs		Non-complete graphs	
			Malicious model (Wang et al., 2022b)	Byzantine model (Bonomi et al., 2019)	Malicious model (Wang et al., 2022b)	Byzantine model
M1 (Buhrman et al., 1995)	Send	–	$n \geq 2F + 1$	$n \geq 3F + 1$	$n \geq 4F + 4$	Open problem
M2 (Garay, 1994)	Any	Yes	$n \geq 3F + 1$	$n \geq 4F + 1$	$n \geq 6F + 4$	Open problem
M3 (Bonnet et al., 2016)	Any	–	$n \geq 4F + 1$	$n \geq 5F + 1$	$n \geq 8F + 4$	Open problem

- The identity of the malicious agents can switch over time. That is, an attacker may turn a non-adversarial agent into a malicious one at certain time instants.
- A malicious agent may recover and become regular. The agent is said to be in the *cured* status at that moment. This happens when the attacker decides to switch to another non-adversarial agent.

Under the mobile adversary model, the identity of the adversaries may switch, but we limit their influence by bounding the total number of them in the network over time. Similarly to our discussion so far, we assume the knowledge of the upper bound F on the total number of such agents. We treat the mobile agents deterministically though the mobile behaviors share similarities with the stochastic models studied for spreading processes of infectious diseases (e.g., Nowzari, Preciado, & Pappas, 2016).

We introduce three classes of mobile malicious behaviors denoted as models M1, M2, and M3. The differences are related to what happens when an adversary moves to another agent and, especially, to whether the recovering agent is aware that it was attacked and its state data may be corrupted. These classes are taken from the literature in computer science for the Byzantine adversaries. We introduce the versions adapted for the malicious adversaries case. As in Algorithm 2.1, in each round, each agent executes three steps: (i) Send its own value, (ii) collect its neighbors' values, and then (iii) update its value. All agents perform the three steps synchronously.

The three classes of mobile malicious models are defined as follows (see Fig. 5).

M1 (Buhrman's model (Buhrman et al., 1995)): The adversary may move away from an attacked agent i only at the sending step in each round k (Fig. 5(b)). At such a round, agent i broadcasts its corrupted state $x_i(k)$ to its neighbors, but then becomes recovered immediately; thus, agent i collects and updates its state as a regular agent. For this reason, agent i will be classified as regular in this round k , i.e., $i \in \mathcal{R}(k)$. If the adversary moved from agent i to another agent j after the send step, then we have $j \in \mathcal{A}(k)$. It is important to note that at each round, there are at most F faulty values in the network.

M2 (Garay's model (Garay, 1994)): This model is characteristic in that each agent has an additional variable, the *cured flag* $\theta_i(k)$; initially, it is set as $\theta_i(0) = 0$. The adversary can move away from an attacked agent i to agent j at any step in each round k (Fig. 5(c)). Then, agent i is classified as adversarial at round k , i.e., $i \in \mathcal{A}(k)$, and as regular in the next round $k + 1$, i.e., $i \in \mathcal{R}(k + 1)$.

In round $k + 1$, agent i is aware that it was infected and sets its flag as $\theta_i(k + 1) = 1$. It is set back to $\theta_i(k + 1) = 0$ after the update step in round $k + 1$. At each round, there are at most F faulty values and F missing values in the network. The cured flag can be used, e.g., to decide whether to make transmissions or not.

M3 (Bonnet's model (Bonnet et al., 2016)): As in the M2 model above, under this model, the adversary agent can move away from an attacked agent i at any step during each round k (Fig. 5(d)). Thus, we have $i \in \mathcal{A}(k)$ and $i \in \mathcal{R}(k + 1)$. At round $k + 1$, agent i is in the recovering state, but is not aware that it was infected. It hence makes the next update as usual. In this case, there are at most $2F$ faulty values in the network: F of them are due to attacks and the remaining F from cured agents like agent i .

It may not be immediate to notice, but these models are numbered in the order of their adversary levels. That is, the M1 model is the most restrictive and easier to deal with while the M3 model represents the most powerful adversary.

Several algorithms to deal with each of these mobile malicious models are provided in Wang et al. (2022b). Here, we briefly overview the approach for the M1 model. The algorithm will be referred to as the modified MSR algorithm for the M1 model. It is mostly the same as Algorithm 2.1 except the deletion step, which is in fact simpler as shown below:

- (Deleting suspicious values in the M1 model): Agent i removes the F largest and F smallest values from its neighbors and its own. The removed data is considered as suspicious and will not be used in the update. The set of the agent indices of the remaining values is written as $\mathcal{M}_i(k) \subset \mathcal{N}_i$.

Note that in this modified MSR algorithm, agent i may remove its own value at any time step k . This is because there is a chance that the agent is not aware of the value being faulty if it just recovered from an attack. While this is a reasonable strategy for maintaining safety, the price to pay is the increased level of robustness in the graph.

We present two results for the resilient consensus under the M1 model from Wang et al. (2022b). The first is for the case of complete graphs, where a tight bound on the number of malicious agents is found.

Theorem 6.2. Consider the multi-agent system whose network $\mathcal{G}$ forms a complete graph. Suppose that the mobile malicious agents follow the F -total and M1 model. Then, the regular agents using Protocol 1 reach resilient consensus if and only if $n \geq 2F + 1$.

Note that the condition $n \geq 2F + 1$ simply indicates that the ratio of the adversarial agents to the regular ones must be less than half. Clearly, this is a hard bound for any agent systems under attacks.

The second result provides a sufficient condition on the network connectivity.

Theorem 6.3. *Consider the multi-agent system under the network $\mathcal{G}$ where the mobile malicious agents follow the F -total and M1 model. Then, the regular agents using the modified MSR algorithm reach resilient consensus if the following conditions are satisfied:*

C1 $n \geq 4F + 4$.

C2 For every agent i , the number of neighbors satisfies $|\mathcal{N}_i| \geq 2F + 1 + n/2$.

Similarly, for the M2 and M3 models, we have results for complete and non-complete graphs in Wang et al. (2022b). Table 2 summarizes the results for the three mobile malicious models. It is interesting that depending on the model of the adversaries, the bound on the number F of adversarial agents with respect to the total number N of agents is different by F . This demonstrates the tradeoff between the network conditions and the adversarial levels of the mobile attacks. In particular, it is interesting that the three models M1, M2, and M3 possess properties with enough differences to make the gap equal to F between the connectivity conditions.

Finally, we remark that the mobile models allow us to consider agents whose statuses are time varying and change between healthy and faulty. The work (Wang, Ishii, Bonnet, & Défago, 2022a) exploited such characteristics to study resilient consensus when infectious diseases are spreading. Agents turn faulty depending on the infection level within the agent community, which is dominated by the popular pandemic disease mathematical model based on SIR (e.g., Nowzari et al., 2016).

6.3. Resilient distributed optimization

We introduce the distributed optimization problem under adversarial agents in Sundaram and Ghahesifard (2018).

The objective is for the agents to solve the global optimization problem in a distributed way:

$$\text{minimize } f(x) = \frac{1}{n} \sum_{i=1}^n f_i(x), \quad (18)$$

where $f_i : \mathbb{R} \rightarrow \mathbb{R}$ is convex with bounded subgradients. We assume that f_i is available only to agent i .

Without adversarial agents, a common approach to solve this problem is to use a synchronous iterative consensus based protocol in which agents use a combination of consensus dynamics and gradient flow to find a minimizer of f . Specifically, each agent sends its estimate to its out-neighbors, receives the estimates of its in-neighbors, and updates its estimate as

$$x_i(k+1) = x_i(k) + \sum_{j \in \mathcal{N}_i} a_{ij}(k) (x_j(k) - x_i(k)) - \alpha_k d'_i(k), \quad (19)$$

where the quantity $d'_i(k)$ is a subgradient of f_i , evaluated at $x_i(k) + \sum_{j \in \mathcal{N}_i} a_{ij}(k) (x_j(k) - x_i(k))$, and α_k is the step-size sequence corresponding to the influence of the subgradient on the update rule at each time step k .

The following result is a special case of the results of Nedic, Ozdaglar, and Parrilo (2010) for graphs that are strongly connected at each time step.

Proposition 6.1. *Suppose the network $\mathcal{G}$ is strongly connected at each time step and that the subgradients of each of the local functions f_i are bounded. Consider the update rule (19). Let the step sizes satisfy $\sum_{k \in \mathbb{Z}_+} \alpha_k = \infty$ and $\sum_{k \in \mathbb{Z}_+} \alpha_k^2 < \infty$. Then, there is a minimizer x^* of (18) such that*

$$\lim_{k \rightarrow \infty} \|x_i(k) - x^*\| = 0 \text{ for } i \in \mathcal{V}.$$

In presence of adversarial agents, the following theorem applies to any algorithm that is guaranteed to output a globally optimum value in the absence of adversaries (even for multivariable functions). The takeaway point is that there is a tradeoff between optimality and resilience: Any algorithm that always finds optimal solutions in the absence of adversaries (under the mild assumptions on the class of local functions) can also be arbitrarily co-opted by an adversary.

Theorem 6.4. *Suppose the local objective functions at each agent are convex with bounded subgradients, but otherwise completely arbitrary. Suppose Γ is a distributed algorithm that guarantees that all agents calculate a global optimizer of problem (18) when there are no adversarial agents. Then, a single adversary can cause all the agents to converge to any arbitrary value when they run algorithm Γ , and furthermore, will remain undetected.*

Theorem 6.4 demonstrates that it is impossible to develop an algorithm that always finds optimal solutions in the absence of adversaries and is at the same time secure against carefully crafted attacks. Based on this fundamental limitations, Sundaram and Ghahesifard (2018) have introduced a secure MSR-like version of the consensus-based distributed optimization protocol:

$$x_i(k+1) = x_i(k) + \sum_{j \in \mathcal{M}_i(k)} a_{ij}(k) (x_j(k) - x_i(k)) - \alpha_k d'_i(k). \quad (20)$$

Note that the MSR local updates in (5) is a specific case of (20) such that $\alpha_k = 0$ for all $k \in \mathbb{Z}_+$.

For the convergence properties of all regular agents $i \in \mathcal{R}$ that follow the dynamics (20), the two theorems below provide sufficient conditions for consensus for scenarios with F -local Byzantine adversaries, and necessary and sufficient conditions for scenarios with F -total malicious adversaries. Let $x_{\mathcal{R}}(k) \in \mathbb{R}^{|\mathcal{R}|}$ be the vector consisting of the states of all regular agents.

Theorem 6.5. *Consider the network $\mathcal{G}$ under the F -local Byzantine model. Suppose that the network is $(2F + 1)$ -robust, the functions f_i for regular agents $i \in \mathcal{R}$, have subgradients bounded by some constant L , and that the regular agents run the dynamics (20) with parameter F . Further, suppose that $\alpha_k \rightarrow 0$ as $k \rightarrow \infty$. Then, there exists a sequence of stochastic vectors $q_k \in \mathbb{R}^{|\mathcal{R}|}$, $k \in \mathbb{Z}_+$, such that*

$$\limsup_{k \rightarrow \infty} \|x_{\mathcal{R}}(k) - \mathbf{1}y(k)\| = 0,$$

$$\text{where } y(k) = q_k^T x_{\mathcal{R}}(k).$$

Theorem 6.6. *Consider the network $\mathcal{G}$ under the F -total malicious model. Suppose that the functions f_i for regular agents $i \in \mathcal{R}$ are convex and have subgradients bounded by some constant L , and that the regular agents run the dynamics (20) with parameter F . Further, suppose that $\alpha_k \rightarrow 0$ as $k \rightarrow \infty$. Then, the regular agents are guaranteed to reach consensus condition $\lim_{k \rightarrow \infty} |x_i(k) - x_j(k)| = 0, \forall i, j \in \mathcal{R}$ if and only if the graph is $(F + 1, F + 1)$ -robust.*

As shown in the previous results, the MSR-like dynamics (20) guarantees consensus within the convex hull of the local minimizers and prevents the adversarial agents from driving the states of the regular agents to arbitrarily large values under the appropriate conditions on the network topology. However, a single malicious agent can still prevent the regular agents from converging to a constant value under certain classes of step sizes.

6.4. More extensions and generalizations

Various extensions and generalizations for realizing resiliency in multi-agent systems and distributed algorithms have been developed. They include but are not limited to those in the following list of recent works:

- Resilient state estimation (Chen, Kar, & Moura, 2018).

- Resilient consensus with input constraints (Wang & Ishii, 2019a).
- Resilient distributed averaging (Dibaji, Safi and Ishii, 2019).
- Resilient consensus with distributed fault detection using multi-hop communication (Yuan & Ishii, 2021b).
- Resilient clock synchronization (Kikuya, Dibaji, & Ishii, 2017).
- Resilient consensus in asynchronous networks (Senejohnny, Sundaram, De Persis, & Tesi, 2019).
- Resilient consensus with privacy protection schemes (Fiore & Russo, 2019).
- Resilient leader-follower consensus (Usevitch & Panagou, 2019).
- Byzantine-resilient distributed optimization (Kuwaranacharoen, Xin, & Sundaram, 2020).
- Byzantine-resilient distributed observers for linear time-invariant systems (Mitra & Sundaram, 2019).

7. Applications for resilient consensus

Consensus is fundamental to various applications in a broad range of engineering areas. In this section, we introduce some typical applications of consensus forming algorithms in wireless sensor networks (WSNs) and robotic networks. We discuss solutions exploiting resilient algorithms to be implemented in environments in the presence of faulty or misbehaving agents.

7.1. Clock synchronization in WSNs

Many applications employing WSNs require highly precise clock synchronization among the sensor agents to maintain consistency in their collected data (Wu, Chaudhari, & Serpedin, 2010). In WSNs, clock synchronization among the sensor agents is necessary because the clocks possess small differences in their speeds. In recent years, fully distributed protocols based on multi-agent consensus techniques have gained much attention (see, e.g., Bolognani, Carli, Lovisari, & Zampieri, 2015; Carli & Zampieri, 2013; Kadowaki & Ishii, 2014; Schenato & Fiorentin, 2011). Under such consensus-based settings, in contrast to more centralized approaches, agents use local information communicated from other agents over the network in order to agree on a common clock value, which is unspecified a priori. The advantage of such protocols lies in their scalability such that agents may easily be added or removed from the network, which raises robustness of the system against failures in sensor nodes and communication.

We briefly present the model of clock synchronization and see how consensus-based protocols may be implemented (Schenato & Fiorentin, 2011). Consider the network of sensor agents connected over the graph $\mathcal{G}$. Let $t \in \mathbb{R}_+$ denote the absolute time, which is not accessible to the agents. Each agent i is equipped with an internal clock whose time $\tau_i^*(t)$ is given as an affine function of t as

$$\tau_i^*(t) = \alpha_i^* t + \beta_i^*, \quad (21)$$

where α_i^* is the clock drift and β_i^* is the clock offset. The values of these parameters may vary within small ranges, which will create differences among the clocks over time. Notice that the true values of the parameters α_i^* and β_i^* are unknown in general and they cannot be modified by the agents either.

Hence, for achieving clock synchronization, each agent i computes and maintains the adjusted time $\tau_i(t)$ based on its own internal time $\tau_i^*(t)$ as

$$\tau_i(t) = \alpha_i(t) \tau_i^*(t) + \beta_i(t), \quad (22)$$

where $\alpha_i(t)$ and $\beta_i(t)$ are, respectively, the new parameters corresponding to the drift and offset. The update laws for these parameters are to be designed. We say that the sensor network achieves clock synchronization if the adjusted clocks of all agents agree asymptotically as

$$\lim_{t \rightarrow \infty} |\tau_i(t) - \tau_j(t)| = 0 \quad \text{for } i, j \in \mathcal{V}. \quad (23)$$

To reformulate this problem as that of consensus, the adjusted time should be rewritten by substituting (21) into (22) as

$$\begin{aligned} \tau_i(t) &= \alpha_i(t) \alpha_i^* t + \alpha_i(t) \beta_i^* + \beta_i(t) \\ &= x_i(t) t + y_i(t), \end{aligned}$$

where the two states, the modified drift $x_i(t) = \alpha_i(t) \alpha_i^*$ and the modified offset $y_i(t) = \alpha_i(t) \beta_i^* + \beta_i(t)$, are introduced. It is now clear that clock synchronization in (23) is achieved if and only if consensus in the two new states are achieved: $|x_i(t) - x_j(t)| \rightarrow 0$ and $|y_i(t) - y_j(t)| \rightarrow 0$ as $t \rightarrow \infty$ for all $i, j \in \mathcal{V}$.

Kikuya et al. (2017) applied the MSR technique to develop a resilient distributed algorithm for clock synchronization in the presence of faulty agents and unreliable communication where packets exchanged may become lost. Specifically, the F -local malicious model is adopted. The agents must run two consensus algorithms for the states $\alpha_i(t)$ and $\beta_i(t)$ for $i \in \mathcal{V}$, both of which can be independently manipulated by malicious adversaries. However, these two states are interdependent in their update laws, and in particular, the error in consensus for the drift $\alpha_i(t)$ affects the consensus process for the offset $\beta_i(t)$.

Based on asynchronous communication among sensor agents, resilient consensus results for the states $\alpha_i(t)$ and $\beta_i(t)$ are established in Kikuya et al. (2017). Without going into the details of the update protocols, they are summarized in the following theorem.

Theorem 7.1. *For the sensor network under the F -local malicious model, suppose that every regular agent i updates its drift $\alpha_i(k)$ and offset $\beta_i(k)$ by the schemes using the MSR algorithm with parameter F . If the underlying network is $(2F + 1)$ -robust, then the sensor network achieves resilient consensus in their modified drifts $x_i(t)$ and modified offsets $y_i(t)$.*

We note that the safety interval S obtained in the above theorem is larger than the typical one in (4). This is because due to the interdependency among the two states mentioned above, decaying noises arise in the overall system. This aspect also has a certain impact on the analysis, making it more complicated.

We further discuss other clock synchronization and desynchronization problems for WSNs in the presence of attacks and failures as follows:

- The work (Phan, Kim, & Kim, 2020) follows the same clock model for WSNs as explained above. It develops an average-based consensus time synchronization protocol with improved robustness in both dynamic and hostile environments, where agents may join or leave the network and unsynchronized agents can consequently bring in adverse effects. In particular, each agent is equipped with a mechanism that keeps track of the status of each of its neighbors, e.g., synchronized, new to the network, or unsynchronized. The agents can then decide whether to include a neighbor in the calculation of the consensus process.
- Another line of research for clock synchronization in WSNs can be found involving the pulse-coupled oscillator model (Hong & Scaglione, 2005; Wang, Nunez, & Doyle, III., 2012). Such a model requires exchange of minimal amount of information among sensor agents, which is the form of pulses. Since the identity of the sender remains anonymous in pulses, clocks following this model may be more vulnerable under malicious attacks. In Wang and Wang (2018), conditions for stealthy attacks are obtained under which synchronization can be guaranteed; however, the network is assumed to have an all-to-all structure, that is, all agents including the malicious ones send pulses to all other agents. In Iori and Ishii (2020, 2021), MSR-based solutions are provided, capable to maintain resiliency for sparse networks with robust graph properties.

- Another application in WSNs is related to desynchronization among transmitters to guarantee equally spaced timing for maximizing throughput at the layer of the Medium Access Control (MAC). Distributed approaches rely on pulse-coupled oscillators and aim at keeping equal spacing among the phase differences. The work (Silvestre, Hespanha, & Silvestre, 2020) develops centralized and distributed algorithms for finding optimal phase offsets for the sensor agents which are resilient to attacks to a subset of agents. The proposed distributed algorithm has limited complexity and performs the detection and isolation of the faulty agent. It also shows a relationship between the variance in the attacker signal and the effects on the agent phases.

7.2. Mobile robot networks

In problems of groups of mobile robots, an important goal is flocking, which refers to the behavior of the robots achieving velocity synchronization and regulation of relative distances within the group (Olfati-Saber, 2006). As an example, we would like to briefly present resilient consensus results applied in resilient flocking from Saulnier, Saldana, Prorok, Pappas, and Kumar (2017).

Consider an n -agent robotic network modeled as a undirected graph $\mathcal{G}$. The location of the robots in the d -dimensional Euclidean space $\mathbb{R}^d$ is denoted by $\mathcal{X} = \{x_1, \dots, x_n\}$ with $x_i \in \mathbb{R}^d$. Their interactions rely on communication following the disk model, where the quality of the link between every pair (i, j) of nodes is based on their locations. Specifically, the quality is expressed by the function $f_{ij} : \mathbb{R}^d \times \mathbb{R}^d \rightarrow [0, 1]$. In Saulnier et al. (2017) and Stump, Jadbabaie, and Kumar (2008), this function is given in the exponential form:

$$f_{ij} = \begin{cases} 1 & \|x_i - x_j\| \leq \underline{C}, \\ 0 & \|x_i - x_j\| \geq \bar{C}, \\ \exp\left(\frac{-\zeta(\|x_i - x_j\| - \underline{C})}{\bar{C} - \underline{C}}\right) & \text{otherwise,} \end{cases} \quad (24)$$

where $\underline{C} > 0$ is the threshold up to which the communication quality is considered optimal, and $\bar{C} > \underline{C}$ is the maximum communication radius. The design parameter $\zeta > 0$ trades off the smoothness of f_{ij} and the magnitude of its derivative. Alternative communication functions can be found in Kim and Mesbahi (2006).

The robots are holonomic, and each robot i moves in the environment based on the control input u_i as

$$\dot{x}_i(t) = u_i(t). \quad (25)$$

When performing consensus, each robot i aims at estimating a global variable of interest y_i ; in the case of flocking, this would be the direction of motion. This goal can be achieved through local interaction, where each robot i updates its own value at time step based on a consensus update rule such as (1). Clearly, this strategy works only when all agents in the network cooperate by executing the update function reliably and communicating truthful values. The models of cooperative and noncooperative robots are given as follows:

Definition 7.1 (Non-Cooperative Robots). A robot is cooperative if it applies the consensus update rule at every time step k and shares the result with its neighbors. It is called non-cooperative otherwise.

Note that such non-cooperative robots can be considered in the framework of Byzantine adversary model. The problem statement for resilient flocking is as follows: Given a networked system of n mobile robots including up to F non-cooperative ones, design a control law that guarantees resilient consensus on the direction of motion. It follows that resilient asymptotic consensus could be guaranteed by MSR-type algorithms with parameter F if the underlying graph is $(2F + 1)$ -robust. Hence, the key problem in resilient flocking is to propose a controller that maintains a certain level of connectivity. This will ensure a resilient

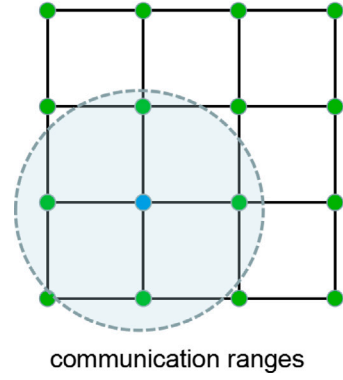


Fig. 6. The network framework on square lattices.

network topology and allowing us to apply the MSR algorithm to compute consensus updates that lead to resilient asymptotic consensus.

To maintain a robust network, Saulnier et al. (2017) proposed a controller based on algebraic connectivity for (25). Note that the algebraic connectivity of the undirected graph $\mathcal{G}$ is the second smallest eigenvalue (in magnitude) of its corresponding Laplacian matrix $L_{\mathcal{G}}$, which we denote by $\mu_{\mathcal{G}}$. The key idea comes from the following theorem:

Theorem 7.2. For a given graph $\mathcal{G}$, it is r -robust with the lower bound

$$r \geq \left\lceil \frac{\mu_{\mathcal{G}}}{2} \right\rceil.$$

An interesting problem in the context of mobile robot teams is the systematic construction of robust graphs. The works such as Guerrero-Bonilla and Kumar (2019), Guerrero-Bonilla, Prorok, and Kumar (2017) and Guerrero-Bonilla, Saldana, and Kumar (2018) proposed such a method by enlarging and modifying a given graph having the robustness property. In particular, the method constructs F -elemental graphs, which are defined as follows:

Definition 7.2 (F -elemental Graph). An F -elemental graph is a graph with $n = 4F + 1$ nodes that is r -robust with $r = 2F + 1$ and a positive integer F .

The method for construction of F -elemental graphs is given in the next proposition.

Proposition 7.1 (F -elemental Graph Construct). Let $\mathcal{V}$ be a set of $n = 4F + 1$ nodes. Assign edges from each of $r - 1 = 2F$ nodes to every other node. Create edges among the remaining $r = 2F + 1$ nodes so that they form a connected subgraph. The resulting graph $\mathcal{G}$ is r -robust.

Another approach for robust graph construction can be designed based on the result in Theorem 3.1. The work (Guerrero-Bonilla et al., 2017) proposes a new method for systematic appending of a subset of an F -elemental r' -robust graph to an r -robust graph. This method ensures that the resulting graph is $\min(r', r)$ -robust.

On the other hand, in Guerrero-Bonilla et al. (2018), the robust graph construction problem is addressed for specific networks. These include connected graphs and those having triangular and square lattice structures. Three main communication ranges $R_{\triangle}$, $R_{\square}$, and $R_{\square}$ are studied, corresponding to formations of robots satisfying the constraints of connected formation, triangular formation, and square formation, respectively. (See the network example of square lattices shown in Fig. 6.) For such specific networks with lattices, r -robustness in a connected formation can be obtained by selecting a proper communication radius for the robots. Let L be the separation distance between robots, then r -robust formations on the connected, triangular and square lattice can be characterized in the result below.

Theorem 7.3. A group of $n \geq 4F + 1$ robots with an associated connected proximity graph is ensured to have a $(2F + 1)$ -robust associated communication graph if the communication range R of each robot satisfies $R \geq 3FL$.

Following this line of research, Guerrero-Bonilla and Kumar (2019) have further presented control laws to drive groups of robots into formations with communication graphs that satisfy the r -robustness property. Using results in r -robustness and control barrier functions, the presented control laws ensure such formations in finite time.

Other resilient coordination problems in mobile robot control include the following:

- Based on a class of MSR algorithms, Renganathan and Summers (2017) provided spoof resilience by incorporating a physical fingerprint analysis of signals received from neighboring agents. By comparing the physical fingerprints of received signals, legitimate agents can identify and isolate malicious agents that attempt spoofing attacks.
- The fault-tolerant algorithms for the problem of gathering n autonomous mobile robots is studied in Agmon and Peleg (2006). An algorithm is provided in this model for gathering $n \geq 3$ robots with at most one faulty robot, and a more general gathering algorithm is given in an n -robot system with up to F faults, where $n \geq 3F + 1$. Moreover, the autonomous mobile robots gathering problem is studied from a probabilistic viewpoint in Défago, Gradinariu, Messika, and Raipin-Parvédy (2006), where a larger set of scheduling strategies such as bounded schedulers is considered, and lower bounds on these schedulers are derived.
- It is shown in Bouzid, Potop-Butucaru, and Tixeuil (2010) that the convergence in robot networks requires specific assumptions about synchrony and Byzantine resilience. Necessary and sufficient conditions for the convergence of mobile robots despite a subset of them being Byzantine is obtained. Additionally, two deterministic convergence algorithms for robot networks are proposed and their correctness and complexity in various atomicity and synchrony settings are analyzed. The resilience of these two algorithms is proved to be optimal.

8. Multi-agent systems under denial-of-service attacks

In this paper so far, our discussions have centered around resilient consensus problems, where adversaries generate false data injection attacks. In this section, we switch the class of attacks to that of DoS attacks and give a general overview on the recent research in the context of multi-agent consensus.

As mentioned in the Introduction, DoS attacks create malicious packet drops in communication channels and hence the availability of data transmitted within the system would be lost. It is obvious that such attacks make it difficult for the agents to achieve consensus, or at least introducing long delays in completing the process. In this section, we mainly focus on the consensus problem of homogeneous multi-agent systems under DoS attacks, which is one of the most fundamental problems.

8.1. Frequency and duration constrained DoS attacks

In this subsection, we introduce the model of DoS attacks, which is based on the characterization of their frequency and duration. It was first introduced by De Persis and Tesi (2015).

This model constrains the attacker's actions by the availability of energy resources for launching DoS attacks. It is clear that if the attacker has sufficient energy to continuously attack, then the agents will not be able to operate properly. On the other hand, if the amount of energy is finite and can be used up, then the agents may simply wait until the attacker terminates after he runs out of energy and then start the consensus forming process. To avoid such trivial situations, this

model assumes that the attacker has a storage of energy which increases at a limited rate over time.

Let $\{h_q\}_{q \in \mathbb{Z}_0}$ with $h_0 \geq \Delta$ denote the sequence of time instants when DoS attacks start, where $\Delta > 0$ denotes the sampling period for the updates and transmissions of the agents. Specifically, $H_q := \{h_q\} \cup [h_q, h_q + \tau_q)$ represents the q th DoS time interval over which the network is in DoS status with length $\tau_q \geq 0$. The DoS may be a single pulse at time h_q when the length is zero, i.e., $\tau_q = 0$.

Given $\tau, t \geq 0$ with $\tau \leq t$, let $n(\tau, t)$ denote the number of DoS attacks launched over the time interval $[\tau, t]$, and let $\Xi(\tau, t) := \bigcup_{q \in \mathbb{Z}_+} H_q \cap [\tau, t]$ be the subset of $[\tau, t]$ where the network is in DoS status. Then, the characterization of DoS attacks is given as follows.

Assumption 8.1 (DoS Frequency). There exist constants $\eta \geq 0$ and $\tau_D > 0$ such that

$$n(\tau, t) \leq \eta + \frac{t - \tau}{\tau_D} \quad (26)$$

for all $\tau, t \geq \Delta$ with $\tau \leq t$.

Assumption 8.2 (DoS Duration). There exist constants $\kappa \geq 0$ and $T > 1$ such that

$$|\Xi(\tau, t)| \leq \kappa + \frac{t - \tau}{T} \quad (27)$$

for all $\tau, t \geq \Delta$ with $\tau \leq t$.

Note that if the attacker compromises all the communication attempts of the multi-agent systems simultaneously, which can be called synchronous DoS attacks, then this type of DoS attacks can be modeled by (26) and (27). If the attacker can launch independent DoS attacks on the communication links between two agents, which can be called asynchronous DoS attacks, then the assumptions regarding DoS frequency and duration need to be generalized. Specifically, for the DoS attacks acting on the communication between agents i and j , the corresponding DoS model should follow $n_{ij}(\tau, t) \leq \eta_{ij} + \frac{t - \tau}{\tau_{ij}^D}$ and $|\Xi_{ij}(\tau, t)| \leq \kappa_{ij} + \frac{t - \tau}{T_{ij}}$ Senejohnny et al. (2017).

8.2. Single integrator multi-agent systems

We start from the basic model of multi-agent systems with agents in the form of single integrators. Such multi-agent systems under DoS were studied in Cetinkaya, Kikuchi, Hayakawa, and Ishii (2020), Senejohnny, Tesi, and De Persis (2015) and Senejohnny et al. (2017).

The agents interact with each other over a network represented by the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each agent i is expressed by the continuous-time system

$$\dot{x}_i(t) = u_i(t), \quad i \in \mathcal{V},$$

where $x_i(t)$ is the state and $u_i(t)$ is the control input.

The controller for computing $u_i(t)$ is a ternary controller with candidate values $-1, 0$, and 1 . The communication is based on requests where each agent i determines the timings for transmissions of requests by the *self-triggered control law*. In particular, at such moments, agent i asks all of its neighbors to send their states and, moreover, determines in advance the instant to send the next request. In the presence of DoS attacks, the states of the neighbors may not reach agent i . Here, the objective of the agents is relaxed to the so-called *practical consensus* in the sense that the states must converge in some finite time $t_f \geq 0$ such that

$$\left| \sum_{j \in \mathcal{N}_i} [x_j(t) - x_i(t)] \right| < \delta \text{ for all } i \in \mathcal{V} \text{ and } t \geq t_f, \quad (28)$$

where $\delta > 0$ is a design parameter representing the desired level of consensus.

We briefly explain the structure of the ternary control. Denote by $\{t'_k\}$ the sequence of time instants when agent i transmits the requests

to its neighbors, where $t_k^i < t_{k+1}^i$ for $k \in \mathbb{Z}_+$. These instants are called transmission attempt times. If agent i successfully hears back from the neighbors at t_k^i , then the input is generated based on the neighbors' information. The control input is immediately applied until some time $\tilde{t}_k^i > t_k^i$ determined by the self-triggering control law and then set its control to zero. Otherwise, it will cease applying control (with $u_i(t) = 0$) until its next transmission time t_{k+1}^i . Specifically, the ternary control input $u_i(t)$ is given by

$$u_i(t) = \begin{cases} \text{sign}_\delta \left(\sum_{j \in \mathcal{N}_i} (x_j(t_k^i) - x_i(t_k^i)) \right) & \text{if no attack at time } t = t_k^i \text{ and } t \in [t_k^i, \tilde{t}_k^i), \\ 0 & \text{otherwise,} \end{cases}$$

where δ is given in (28) and

$$\text{sign}_\delta(u) = \begin{cases} \text{sign}(u) & \text{if } |u| \geq \delta, \\ 0 & \text{otherwise.} \end{cases}$$

The papers by Cetinkaya et al. (2020) and Senejohnny et al. (2015) consider the scenario where the attacker can compromise all the communication among the agents simultaneously when he attacks the system. In Senejohnny et al. (2015), the multi-agent systems can achieve practical consensus in (28) if DoS attacks satisfy $\frac{1}{T} + \frac{\Delta_*}{\tau_D} < 1$, where Δ_* is the minimum interval between transmission attempt times. There, the transmission attempt times are deterministically chosen. Thus, if the attacker knows how they are chosen and is moreover capable to launch sufficiently frequent attacks, he may disable all communication without spending much energy. This can be done by generating very short attacks exactly at the transmission attempt times t_k^i .

The work (Cetinkaya et al., 2020) demonstrates that a randomization-based approach can enhance the system robustness. In particular, it is shown that by randomly selecting the transmission attempt times $\{t_k^i\}$, the attacker cannot block all communication even by frequent DoS attacks. In this case, the agents can arrive at practical consensus (28) under the more relaxed condition of $\frac{1}{T} < 1$. It is interesting to note that this is another instant where a randomization-based strategy can be effective in mitigating the influence of attacks and serves as a protective measure. Recall the discussion in Section 5.2 on quantized resilient consensus.

8.3. General multi-agent systems under DoS attacks

In this subsection, we proceed to study multi-agent systems under DoS attacks, where the agents have more general linear dynamics in the continuous-time domain.

As in the previous case, consider a group of agents communicating over the network represented by the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. For each agent i , its state–space equation is given by

$$\dot{x}_i(t) = Ax_i(t) + Bu_i(t), \quad (29)$$

where $i \in \mathcal{V}$, $x_i(t) \in \mathbb{R}^n$ is the state and $u_i(t)$ is the control input. We assume that the matrices A and B have compatible dimensions. The consensus is defined in terms of the whole states of the agents as

$$\lim_{t \rightarrow \infty} \|x_i(t) - x_j(t)\| = 0 \quad \text{for } i, j \in \mathcal{V}.$$

For achieving consensus, communication is necessary for the computation of the control input $u_i(t)$. Agent i can only exchange information with its neighbor agents $j \in \mathcal{N}_i$. For the multi-agent system in (29), with the application of event-triggered control, the paper (Feng & Hu, 2020) provides the design of controller given in the form

$$u_i(t) = K \sum_{j=1}^N a_{ij}(\hat{x}_j(t) - \hat{x}_i(t)), \quad (30)$$

where K is a feedback gain matrix. In particular, $\hat{x}_j(t)$ with $j \in \mathcal{V}$ in $u_i(t)$ is computed as

$$\begin{cases} \dot{\hat{x}}_j(t) = A\hat{x}_j(t) & \text{if } t \in [t_k^j, t_{k+1}^j), \\ \hat{x}_j(t) = x_j(t) & \text{if } t = t_k^j, \end{cases} \quad (31)$$

where t_k^j and t_{k+1}^j denote consecutive successful transmission instants of agent j , during which DoS attacks may be present. Observe that between two successful transmissions (possibly DoS can occur in between), agent i estimates the state of agent j based on the first equation in (31). If agent i has a successful transmission from agent j , then agent i updates the estimate $\hat{x}_j$ of its state as $\hat{x}_j(t) = x_j(t_k^j)$. It is shown that if the attack frequency and duration are upper bounded by certain values (depending on system matrices and the graph structure), then consensus can be achieved. Moreover, through the use of self-triggered control, Zeno behaviors (i.e., very frequent transmissions) that often cause issues in event-triggered control can be prevented.

Up to now, we have discussed resilient consensus for the specific case where the agents are homogeneous in their dynamics. In the literature, the consensus problem under DoS is also considered from various directions. The output leader–follower consensus problem of heterogeneous multi-agent systems under DoS attacks is considered in Zhang, Liu, and Feng (2018), where the multi-agent systems are subject to aperiodic sampled transmissions. In particular, the DoS attacks are characterized by an i.i.d. Bernoulli process. In Tang et al. (2020), the problem of formation control of nonlinear multi-agent systems in the presence of DoS attacks is investigated. Lu and Yang (2018) considered asynchronous DoS attacks, i.e., the attacker is able to launch independent DoS attacks on different communication links. The work (Liu & Wang, 2021) also studies asynchronous DoS attacks in which the problem is transformed into a two-layer switched system with one stable mode and several unstable modes. Deng and Wen (2020) comprehensively considered the factors mentioned above, i.e., heterogeneous multi-agent systems and asynchronous attacks, and additionally consider actuator faults. A resilient fault-tolerant control problem under DoS attacks is formulated, and the control objective is to achieve output regulation. Novel resilient adaptive distributed observers are designed in order to maximize the resilience against DoS, i.e., maximizing the tolerable DoS duration. The recent paper by Nugraha, Cetinkaya, Hayakawa, Ishii, and Zhu (2021) considers multi-agent systems under jamming attacks that can target individual links by formulating a game between the defender and the attacker. The attacker attempts to remove some of the communication links between two agents and the defender attempts to recover the links.

8.4. Quantized consensus of multi-agent systems under DoS attacks

In the previous subsection, one implicit assumption made about the communication within the multi-agent systems is that the bandwidths in the channels are infinite. This implies that the communicated signals can have infinite precision. However, due to the nature of digital communication, signals to be transmitted must go through analog-to-digital conversion and hence will be quantized. In this subsection, we introduce some recent works of consensus problems under DoS attacks in which the transmitted signals of multi-agent systems are subject to quantization.

Quantized consensus problems of multi-agent systems have been broadly studied in the last decade (Carli et al., 2010; Kashyap et al., 2007; Qiu, Xie, & Hong, 2015; Wang & Ishii, 2020; You & Xie, 2011) and some of them take data rate constraints into considerations. Indeed, the issue of data rate in networked control of multi-agent systems will become relevant especially if the agents try to reach consensus on multiple variables and the volume of data communicated among agents is large. Reducing the data size for each variable is one way to make the system robust to changes in available data rates in, e.g., wireless communication.

Though the problems of quantized consensus have been widely investigated, the ones under DoS attacks are still relatively new. The recent paper (Amini, Asif, & Mohammadi, 2021) considers such problems using the DoS attack model characterized by the DoS frequency and duration from (26) and (27). Note that quantization employed in the work has infinite range. This simplifies the problem since there is

no problem regarding quantizer overflow, and moreover quantization error is always bounded by a fixed constant.

To consider constraints of *finite data rates* in communication channels, we must take account of quantization mechanisms with finite range. One such quantization mechanism called *dynamic quantization* adjusts the quantization range dynamically adjusted along the system state evolutions (Ishii & Francis, 2002; Nair & Evans, 2004; Tatikonda & Mitter, 2004). Such a dynamic quantization mechanism has also been extended to the case of multi-agent systems without packet losses in, e.g., You and Xie (2011). However, there are few works simultaneously addressing the two issues of dynamic quantization and consensus in the presence of packet losses, not to mention DoS attacks.

In what follows, we present the problem settings of the recent work by Feng and Ishii (2022). It investigates the use of dynamic quantization in consensus under DoS attacks, accounting for the finite data rate in communication among agents. The agents interacting over the network $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ are expressed as homogeneous general linear systems $x_i(k) = Ax_i(k-1) + Bu_i(k-1)$ where $k \in \mathbb{Z}_+$, $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times w}$. It is assumed that (A, B) is stabilizable. The vector $x_i(k) \in \mathbb{R}^n$ denotes the state of agent i . The communication attempts take place periodically along k . Let $u_i(k-1) \in \mathbb{R}^w$ denote its control input, whose computation will be given later. The control objective is state consensus for $x_i(k)$ as $\lim_{k \rightarrow \infty} \|x_i(k) - \frac{1}{N} \sum_{j=1}^N x_j(k)\| = 0$, for $i \in \mathcal{V}$.

For each agent i , the control input $u_i(k)$ is expressed as a function of the relative states available locally at time k . Specifically, it is given by $u_i(k) = K \sum_{j=1}^N a_{ij}(\hat{x}_j(k) - \hat{x}_i(k))$ and $\hat{x}_p(k) \in \mathbb{R}^n$ ($p = i, j$) denotes the estimation. The matrix K is an appropriate feedback gain matrix, which can ensure state consensus if there are no DoS attacks.

Agent i estimates the states of its neighbors based on the information available from communication. Also, to stay consistent with the neighbors, it will compute the estimate of its own. These estimated states will be updated at each time as

$$\hat{x}_j(k) = \begin{cases} A\hat{x}_j(k-1) + \theta(k-1)\hat{Q}_j(k) & \text{if } k \notin H_q, \\ A\hat{x}_j(k-1) & \text{if } k \in H_q, \end{cases} \quad (32)$$

where $j \in \{i\} \cup \mathcal{N}_i$ and the initial estimates will be set as $\hat{x}_j(0) = 0$. The scaling parameter $\theta(k) \in \mathbb{R}_+$ in (32) is updated as

$$\theta(k) = \begin{cases} \gamma_1 \theta(k-1) & \text{if } k \notin H_q, \\ \gamma_2 \theta(k-1) & \text{if } k \in H_q \end{cases} \quad (33)$$

with $\theta(0) = \theta_0 \in \mathbb{R}_+$, where $0 < \gamma_1 < 1$ and $\gamma_2 > 0$.

Moreover, the scaling parameter $\theta(k)$ is used in the quantization $\hat{Q}_j(k)$ given by

$$\hat{Q}_j(k) = Q_R \left(\frac{x_j(k) - A\hat{x}_j(k-1)}{\theta(k-1)} \right) \quad (34)$$

for preventing quantizer overflow. The function $Q_R(\cdot)$ represents the quantization effects introduced by the uniform quantizer. By adjusting the size of $\theta(k)$ dynamically, the state will be kept within the bounded quantization range without saturation, i.e., $\frac{x_j(k) - A\hat{x}_j(k-1)}{\theta(k-1)}$ in $Q_R(\cdot)$ is upper bounded by some certain values. The parameters γ_1 and γ_2 in (33) are for zooming in and out such that the quantization scaling parameter $\theta(k)$ changes dynamically both to mitigate the influence of DoS attacks and to reach asymptotic consensus using finite bit rate. Under DoS attacks, the states of the agents to be quantized may diverge. Therefore, the quantizers must zoom out and increase their ranges so that the states can be measured properly. If the transmissions succeed, the quantizers zoom in and $\theta(k)$ decreases by using γ_1 .

A tradeoff characterization between resilience of the agent system to DoS attacks and the necessary data rate in communication is also given (Feng & Ishii, 2022). For improving the robustness, it helps to use small γ_1 and γ_2 , which will enlarge the class of tolerable DoS attacks. On the other hand, a small γ_1 will result in large data rate. Intuitively, this tradeoff has a clear implication: Higher resilience needs more data rate.

Different from the distributed control law, the paper (Yang, Ju, Xia, & Zhang, 2021) applies a cloud-based control scheme for the control of multi-agent systems, in which communication time delays and quantization are both taken into consideration. The control inputs of the multi-agent systems are computed in the cloud. The DoS attacks are characterized by the maximum consecutive packet losses, i.e., the number of packet drops in a series induced by the attackers has an upper bound.

9. Conclusion

In this paper, we have reviewed some recent developments on a range of problems related to resilient consensus in multi-agent systems under the influence of malicious adversaries. We have highlighted the active research in this new interdisciplinary area. Then, we have placed special emphases on the particular classes of attacks categorized into the two types.

The first class is that of data injection attacks, where part of the agents in a network may be considered being adversarial. We have given a thorough overview on the interesting approach based on MSR algorithms and their variants. The basic problem setting as well as the algorithm structure have been discussed. We have seen their effectiveness for enhancing resiliency in reaching consensus against both malicious and Byzantine type adversaries. In particular, theoretical guarantees can be established when the number of adversarial agents is limited and the network possesses sufficient structural properties in terms of connectivities. It was then observed that MSR-type algorithms can be useful for agent networks with further dynamics as well as those for distributed optimization. Several applications have also been discussed such as clock synchronization of WSNs and mobile robot networks.

The second class of adversaries consists of DoS attacks and jamming attacks, which can block proper communication among the agents within the network. An attack model characterized by energy constraints on the attackers' side has been discussed. We have gone over analysis approaches towards finding bounds on the attack levels in terms of frequencies and durations to guarantee consensus for different agent problems. Different from the probabilistic packet drop models, these studies are carried out deterministically.

Finally, it is evident that the importance of security for cyber-physical systems will continue to grow, even at a faster speed, especially in view of the increasing number of serious attack incidents in applications. Under such situations, it is vital to extend fundamental research on various measures for securing control and measurement data in distributed control systems including multi-agent systems. For problems related to resilient consensus and agent systems under DoS attacks, we would like to mention below a few directions for future research:

1. As seen in this paper, robust graphs play key roles in the problem of resilient consensus, but checking the degree of robustness of given graphs can be computationally prohibitive for large networks. More efficient algorithms for verification as well as construction of robust networks should be studied.
2. One general principle of MSR algorithms is that detection of adversaries is not performed by the individual agents. This aspect can be strengthened by keeping records of the behaviors of neighbors and potentially applying learning-based techniques. Such studies may lead us to finding adversaries capable to escape such detection mechanisms, which will require development of algorithms with higher resilience.
3. We have presented several consensus problems involving agents with higher-order dynamics, but this direction can be further pursued for other applications in networked control. In this type of research, we can take advantage of the systems control approach, which might facilitate collaboration with researchers in other domains.

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